

1 Three-component decomposition of treatment response
2 in aggregated N-of-1 trials: pharmacological,
3 expectation-driven, and natural-history components

4 pmsimstats team

5 2026-06-15

6 **Contents**

7	Abstract	2
8	1 Introduction	3
9	1.1 A claim about how N-of-1 trials should be analysed	3
10	1.2 Why this matters: three failure modes of the lumped analysis . . .	4
11	1.3 The current state of the N-of-1 literature	8
12	1.4 The neurobiological and epidemiological reality of the components	11
13	1.5 What this paper contributes	13
14	2 Notation and the formal model	13
15	3 The three components	16
16	3.1 Biological response (BR)	16
17	3.2 Placebo-belief response (PB)	20
18	3.3 Time-variant natural-history component (TV)	29
19	4 Identifiability and trial-design contrasts	34
20	5 Analysis-side methodology	37
21	5.1 Four reasons not to match the DGP	38
22	5.2 A small extension that usually pays off: phase indicators	41
23	5.3 When to fit the full decomposition	42
24	6 Worked example: information loss without decomposition	44
25	6.1 What the lumped analysis estimates	44
26	6.2 A numerical illustration	47
27	7 Simulation study	53
28	7.1 Pilot validation of analysis machinery (100 replicates, 6 cells) . . .	53
29	7.2 Production-simulation specification	55

30	7.3 Study A results: bias and power at 1,000 replicates per alternative	
31	cell	56
32	8 Application to predictive-biomarker validation	58
33	9 Discussion	61
34	10 Conclusions	65
35	11 Conflict of interest	67
36	12 Funding	67
37	13 Data availability statement	67
38	14 Reproducibility	67
39	15 References	68

40 Abstract

41 **Background.** Treatment response in aggregated N-of-1 clinical trials reflects
42 three causally distinct mechanisms: pharmacological action of the active com-
43 pound (BR), the patient’s belief that the treatment will work (PB), and natural-
44 history drift in the underlying condition (TV). Standard analysis models lump
45 these into a single ‘treatment effect’ plus residual error. Whether this lumping
46 biases inference is not unconditional: it depends on the estimand and on whether
47 candidate biomarkers correlate with the non-pharmacological components, a de-
48 pendence this paper makes precise.

49 **Methods.** We formalise a three-component decomposition of the within-
50 participant outcome trajectory, each component a modified Gompertz function
51 with participant-specific random effects, and characterise the trial-design con-
52 trasts (open-label, blinded discontinuation, crossover) that supply identifiability.
53 We derive the omitted-variable-bias identity for the one-component estimator,
54 which expresses both the lumped treatment effect and the lumped biomarker
55 slope as sums of component-specific terms, and we evaluate the analysis strategies
56 in a pre-registered Monte Carlo study (Study A) calibrated to the prazosin-PTSD
57 reference parameters, comparing a one-component design-contrast analysis with
58 a phase-augmented analysis at 1,000 replicates per alternative cell (5,000 per null
59 cell) across $m_{PB} \in \{0, 1, 3, 6, 10\}$, $m_{TV} \in \{-1, 0, 1, 2\}$, and $N \in \{35, 70, 100, 150\}$.

60 **Results.** The identity shows that the lumped biomarker-by-treatment slope is dis-
61 placed from the pharmacological slope by a term proportional to the biomarker’s
62 covariance with PB and TV, not by the magnitude of those components. Study
63 A confirms this: with a biomarker coupled to BR only, the one-component esti-
64 mate of the biomarker-by-treatment interaction is essentially unbiased across the
65 whole grid (mean absolute bias 0.016 at $N = 35$ and 0.006 at $N = 150$, within one

Monte Carlo standard error of zero), with near-nominal coverage and type I error. The phase-augmented analysis provides no bias reduction and is markedly less powerful (mean power 0.35 versus 0.59 at $N = 35$; 0.62 versus 0.90 at $N = 70$), so it is dominated under this uncontaminated biomarker. The inferential value of decomposition is therefore conditional, not universal.

Conclusions. The trial design must supply the drug-on/off and belief contrasts; given those, component-level decomposition is useful when the estimand is the attribution of response to pharmacology versus belief versus natural history, or when a candidate biomarker may correlate with PB or TV. For a biomarker-by-treatment interaction with a biomarker orthogonal to PB and TV, a single-indicator analysis that exploits the design contrast is sufficient and a parametric component model is counterproductive. A pragmatic specification for the cases that warrant it is $Sx \sim bm + t + Dbc + phase + Dbc:phase + bm:Dbc + bm:Dbc:phase$ with random intercept and AR(1) residual correlation. The decisive test of the biomarker-validation failure, a data-generating process that couples the biomarker to PB, remains to be run.

1 Introduction

1.1 A claim about how N-of-1 trials should be analysed

[1]

This paper argues that treatment response in N-of-1 trials comprises three causally distinct components that must be analysed jointly rather than lumped.

- The three components are *BR* (pharmacological), *PB* (placebo-belief), and *TV* (natural history).
- Each component is causally distinct, clinically interpretable, and identifiable from a well-designed N-of-1 trial.
- The standard single-indicator mixed-effects model conflates all three and renders biomarker validation unanswerable.

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.

This paper advances an opinionated methodological claim: treatment response in aggregated N-of-1 clinical trials is the sum of three causally distinct components and should routinely be analysed as such, rather than estimated as a single lumped quantity. The three components are pharmacological action (*BR*, the biological response), placebo-belief response (*PB*, the patient's belief that they are receiving treatment), and time-variant natural history (*TV*, the trajectory that would have occurred without intervention). Each is causally distinct, each is independently of clinical and regulatory interest, and each is identifiable from a properly designed N-of-1 trial. We shall argue that the standard

106 practice of fitting a single mixed-effects model with one treatment indi-
107 cator, prevalent across the published N-of-1 literature, conflates these
108 three components, biases the interpretation of treatment effect, and
109 renders predictive-biomarker validation unanswerable as a scientific
110 question.

111 [2]

112 **The contribution is the consolidation and extension of an**
113 **established structural-decomposition tradition, not a novel**
114 **decomposition.**

- 115 • The additive disease-progression, placebo, and drug-action de-
116 composition is well established in pharmacometrics and longitu-
117 dinal trial methodology.
- 118 • The present framework extends that tradition with an explicit,
119 separately identified placebo-belief channel and brings it to the
120 aggregated N-of-1 setting.
- 121 • The decomposition logic is not specific to N-of-1; it applies wher-
122 ever a longitudinal treatment response is analysed.

123 *Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod*
124 *tempor incididunt ut labore et dolore magna aliqua.*

125 We make no claim of novelty for the decomposition itself. The struc-
126 tural decomposition of a longitudinal treatment response into sepa-
127 rable disease-progression, placebo, and drug-action components is a
128 well-established tradition in pharmacometrics^{1–3}, and the separation
129 of drug from placebo response in longitudinal trajectories has been
130 pursued both through growth-mixture models⁴ and through experi-
131 mental decomposition of the placebo response itself⁵. Each ingredient
132 we use (mixed-effects models with trial-phase indicators, placebo-arm
133 contrasts, regression-to-mean adjustment, growth-curve parameterisa-
134 tion) has likewise appeared across decades of clinical-trial methodol-
135 ogy. Our contribution is therefore one of consolidation and extension
136 rather than invention: we make the three-component structure explicit
137 for the aggregated N-of-1 setting, add a separately modelled placebo-
138 belief channel identified by design contrasts rather than by functional
139 form alone, and argue that the decomposition should become routine
140 wherever a longitudinal treatment response is analysed. We document
141 why the framing has not been routine in N-of-1 practice (parsimony,
142 identifiability concerns, sample-size constraints, methodological iner-
143 tia) and propose an analysis that is tractable at trial-relevant sample
144 sizes through a phase-augmented linear-mixed-effects approximation
145 to the full decomposition.

146 1.2 Why this matters: three failure modes of the lumped analysis

147 [3]

148 **The lumped one-component analysis has three documented**
149 **failure modes that grow with unmodelled component hetero-**
150 **geneity.**

- 151 • The three failure modes are placebo-attribution bias, regression-
152 to-the-mean, and biomarker-validation failure.
- 153 • Each failure mode is documented in the chronic-condition trial
154 literature, not merely hypothetical.
- 155 • The failures are present regardless of sample size.

156 *Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod*
157 *tempor incididunt ut labore et dolore magna aliqua.*

158 We begin with the failure modes of the standard approach. A one-
159 component analysis (‘the treatment effect is what we observe minus
160 baseline’) has three identifiable failure modes that arise when the pop-
161 ulation displays meaningful heterogeneity in any of the unmodelled
162 components. None of these is hypothetical: each has a documented
163 presence in the chronic-condition trial literature. Two points of preci-
164 sion, established formally in section 6.1 and borne out by the simula-
165 tion of section 7.3, should be carried through what follows. The first
166 two failure modes attach to the baseline-anchored lumped effect and
167 are removed once the design supplies a drug-on/off contrast, which
168 the one-component analysis of later sections in fact uses. The third,
169 biomarker-validation failure, is different in kind: it requires not merely
170 that PB or TV be large but that the candidate biomarker correlate
171 with them, and where that coupling is absent the lumped analysis is
172 not biased at all.

173 [4]

174 **Placebo-attribution bias is sample-size-invariant and docu-**
175 **mented across antidepressant, anxiety, and IBS literatures.**

- 176 • Lumped treatment-effect estimates overstate pharmacological ef-
177 ficacy in proportion to PB magnitude and variance.
- 178 • Kirsch et al. and Locher et al. demonstrated that antidepressant
179 apparent efficacy is largely explained by placebo response and
180 natural fluctuation.
- 181 • Placebo responsiveness has measurable biological correlates (e.g.
182 COMT genotype), confirming it is a structured phenomenon, not
183 noise.

184 *Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod*
185 *tempor incididunt ut labore et dolore magna aliqua.*

186 The first is the placebo-attribution failure. When the population con-
187 tains both high and low placebo responders, the lumped treatment
188 effect overestimates the pharmacological component. The bias scales
189 with the magnitude and variance of PB in the study population, and

190 it is invariant under sample-size increases: a larger one-component
191 trial converges on the same biased estimate. The most consequential
192 example in modern clinical pharmacology is antidepressant efficacy.⁶
193 analysed FDA-submitted data on four widely- prescribed antidepres-
194 sants and showed that, except at the most severe baseline-severity
195 stratum, mean drug-placebo differences fell below clinical-significance
196 thresholds, with the apparent ‘efficacy’ largely explained by placebo
197 response and natural fluctuation;⁷ reproduced this pattern in late-life
198 depression and explicitly invoked regression to the mean.⁸ generalised
199 the finding to anxiety. Patient populations differ systematically in
200 placebo responsiveness, and the difference has measurable biological
201 correlates: COMT val158met genotype predicts placebo response in
202 irritable bowel syndrome^{9,10}, indicating that ‘placebo’ is not a homoge-
203 neous nuisance but a structured biological phenomenon that interacts
204 with the patient’s genome and treatment history. A trial whose pop-
205 ulation happens to be placebo-responsive will report a larger ‘drug
206 effect’ than one whose population is not, even when the underlying
207 pharmacology is identical.

208 [5]

209 **Regression-to-the-mean contributes several points of appar-**
210 **ent improvement in early trial weeks and is absorbed without**
211 **trace by the lumped analysis.**

- 212 • Enrolment at peak severity guarantees a natural-history regres-
213 sion component irrespective of pharmacology.
- 214 • Hrobjartsson et al. systematic reviews across 200+ trials showed
215 much of conventional placebo response was indistinguishable
216 from natural-history drift.
- 217 • The lumped analysis has no formal mechanism to recover the
218 pharmacological effect once regression is absorbed.

219 *Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod*
220 *tempor incididunt ut labore et dolore magna aliqua.*

221 The second failure mode is regression-to-the-mean. When patients en-
222 rol at moments of clinical engagement that coincide with peak symp-
223 tom severity, natural-history regression toward typical severity con-
224 tributes several points of apparent improvement in the early weeks
225 of any trial, without any pharmacological contribution at all. The
226 systematic reviews of^{11, 12}, and¹³, the last covering 202 trials across 60
227 clinical conditions, quantified this directly: in trials with both placebo
228 and no-treatment arms, much of what is conventionally attributed to
229 the placebo response was indistinguishable from regression to the mean
230 and natural-history drift. The lumped analysis absorbs this regression
231 into the treatment-effect estimate and the analyst has no formal mech-
232 anism to recover the true pharmacological effect.

Biomarker validation is invalidated when a biomarker correlates with PB or TV , rendering precision-medicine stratification wrong regardless of trial size.

- A biomarker correlated with placebo responsiveness or natural-history trajectory is indistinguishable from a true pharmacological predictor in a lumped analysis.
- The PATH statement defers mechanistic attribution to domain knowledge; the BR-PB-TV decomposition supplies that substrate.
- Where such coupling cannot be ruled out, rigorous pharmacological-biomarker validation requires the decomposition; where the biomarker is known orthogonal to PB and TV , the lumped analysis already recovers the pharmacological slope.

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.

The third failure mode is the biomarker-validation failure. The precision-medicine question that motivates many recent N-of-1 trials is whether a baseline biomarker predicts treatment response, formalised in the PATH statement^{14,15}. A biomarker that correlates with placebo responsiveness or with a favourable natural-history trajectory will appear in a one-component analysis as a predictor of the lumped ‘treatment effect’, indistinguishable from a true pharmacological predictor. The biomarker-stratified prescribing decision that follows from such an analysis is wrong, and the trial’s regulatory utility is degraded. The PATH framework explicitly defers the mechanism of treatment-effect heterogeneity to domain knowledge^{16,17}; the BR-PB-TV decomposition supplies the mechanistic substrate that PATH leaves blank. We argue that, whenever the candidate biomarker may itself correlate with PB or TV , predictive-biomarker validation cannot be conducted rigorously in a lumped-analysis framework regardless of how large the trial is; the bias is fixed by the biomarker’s coupling to the non-pharmacological components, not by sample size. Where the biomarker can be assumed orthogonal to PB and TV , this failure mode does not arise, and the lumped analysis recovers the pharmacological slope, as the simulation of section 7.3 confirms. We stress at the outset that this failure mode is established in this paper as an algebraic identity (section 6.1), not yet as a simulated phenomenon: the executed Study A (section 7.3) deliberately varies the *magnitude* of PB and TV with the biomarker coupled to BR alone, which the identity predicts to be innocuous, while the decisive cell that couples the biomarker to PB remains to be run. The failure-mode claims of this section should therefore be

276 read as consequences of the identity, whose empirical demonstration
277 is a stated item of outstanding work rather than a result reported
278 here.

279 1.3 The current state of the N-of-1 literature

280 [7]

281 **The N-of-1 literature has matured substantially but moti-**
282 **vates three observations that the present paper addresses.**

- 283 • The design has evolved from individual-patient consultation to a
284 framework supporting population-level inference.
- 285 • CONSORT reporting standards and quality audits document pro-
286 gressive improvement in analytic transparency.
- 287 • Three specific gaps in the literature motivate the decomposition
288 framework proposed here.

289 *Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod*
290 *tempor incididunt ut labore et dolore magna aliqua.*

291 The N-of-1 trial in its modern randomised form originates with¹⁸, who
292 defined the within-patient multiple-crossover RCT, demonstrated it on
293 a single airflow-limitation case, and established a clinical service to de-
294 liver it at scale. The design inherits its identifying logic from two
295 older traditions: the crossover trial of classical biostatistics, in which
296 within-subject period contrasts estimate treatment effects under ex-
297 plicit carryover assumptions, and the single-case experimental designs
298 of behavioural science, in which phase reversals (withdrawal and rein-
299 statement of treatment) identify effects within one participant. The
300 blinded-discontinuation contrast central to this paper is a direct de-
301 scendant of both. The methodological step from the individual-patient
302 trial to population-level inference was taken by¹⁹ and²⁰, who showed
303 that a series of single-patient trials can be combined through hierarchi-
304 cal meta-analysis to estimate both population treatment effects and
305 individual responses, the statistical foundation on which the aggre-
306 gated design rests. Over the last fifteen years the design has accord-
307 ingly been reframed from an individual-patient decision tool into an
308 aggregated, meta-analytic framework supporting population-level in-
309 ference: the programmatic statement of²¹, the methodological reviews
310 of²², the historical-practice audit of²³, and the more recent syntheses
311 of²⁴,²⁵,²⁶, and²⁷ trace that trajectory. Reporting standards are codi-
312 fied in the CONSORT extension²⁸ (the headline statement) and²⁹ (the
313 explanation-and-elaboration companion); the reporting-quality audits
314 of³⁰ and its successors document slow but progressive improvement in
315 analytic transparency. Three observations about this literature moti-
316 vate the present paper.

317 [8]

318 **No published N-of-1 trial employs a structural decomposition**
319 **of treatment response despite three decades of methodologi-**
320 **cal literature.**

- 321 • Audit data show the field has relied on paired t-tests, binary-
322 indicator LME, and hierarchical Bayesian regressions.
- 323 • Recent methodological advances (Liao 2023, Blackston 2019, Car-
324 rozzo 2025) retain the lumped-response framing.
- 325 • The decomposition gap persists even as carryover and identifica-
326 bility problems are increasingly recognised.

327 *Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod*
328 *tempor incididunt ut labore et dolore magna aliqua.*

329 The first observation is that the dominant analytic models are sim-
330 ple. The reporting-quality audit of³⁰ documents that the published
331 N-of-1 trials between 1985 and 2013 relied predominantly on paired
332 t-tests, simple linear mixed-effects models with a binary treatment in-
333 dicator, or hierarchical Bayesian regressions. On our own reading of
334 that corpus and the methodological literature that followed it, no pub-
335 lished N-of-1 trial has employed a structural decomposition of treat-
336 ment response into pharmacological, expectancy, and natural-history
337 components. The most recent methodological advances retain this
338 character:³¹ develops Bayesian distributed-lag models with autocor-
339 related errors as a current carryover-aware analysis, but the model
340 still treats the response as a single quantity to be estimated.³² and³³
341 compare aggregated N-of-1 trials with parallel and crossover RCT al-
342 ternatives by simulation and document carryover-induced type-I-error
343 inflation as a key failure mode, but again at the level of the lumped
344 treatment effect. None of these papers decompose the response.

345 [9]

346 **Placebo-component separation is widely recognised but al-**
347 **most entirely unaddressed at the protocol level in N-of-1**
348 **methodology.**

- 349 • The Hendrickson hybrid blinded-discontinuation phase is among
350 the few protocol mechanisms that supply *PB*-versus-*BR* identi-
351 fiability.
- 352 • Placebo run-in, open-label-placebo, and mediator-analysis
353 approaches have not been imported into N-of-1 designs.
- 354 • The gap between awareness of the problem and protocol-level
355 solutions motivates the present framework.

356 *Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod*
357 *tempor incididunt ut labore et dolore magna aliqua.*

358 The second observation is that the placebo-component separation
359 problem is widely known but largely unaddressed in N-of-1 specifically.

The blinded discontinuation phase used in the Hendrickson hybrid design³⁴ is one of the few protocol-level mechanisms in routine N-of-1 use that supplies identifiability of *PB* versus *BR*. Other relevant approaches, including placebo run-in designs, open-label-placebo trials, and mediator-analysis applications, have been developed in parallel but rarely imported into N-of-1 methodology.

[10]

Kaptchuk’s within-placebo decomposition is the closest structural analogue; no published paper applies the corresponding full-response decomposition to N-of-1 trials.

- Kaptchuk et al. (2008) empirically decomposed the placebo response into observation, ritual, and relationship components.
- Open-label placebo trials confirm that clinically meaningful PB-only response is possible even when patients know they receive placebo.
- The BR-PB-TV framework is structurally analogous but operates at the level of the full treatment response.

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.

The third observation is that the closest precedent in the literature is Kaptchuk’s three-component decomposition of the placebo response itself. In a randomised trial of acupuncture-style placebo for irritable bowel syndrome,⁵ decomposed the placebo response into observation, ritual, and patient-practitioner relationship components, demonstrating empirically that the placebo response is itself a sum of identifiable parts rather than a single ‘expectancy’ lump. The follow-up open-label placebo trials of^{35, 36}, and³⁷ in adolescent functional pain establish that even disclosed-placebo intervention can produce clinically meaningful PB-only response. The framework we propose is structurally analogous to Kaptchuk’s decomposition but applied at the level of the full treatment response (drug, placebo, natural history) rather than within the placebo response alone. To the best of our knowledge, no published paper proposes the corresponding decomposition for the aggregated N-of-1 setting, even though the methodological ingredients, including mixed-effects modelling, blinded-phase contrasts, and trajectory parameterisation, are all in routine use.

[11]

The decomposition gap is real, the tools to fill it are mature, and the failure modes are large enough to make it a methodological priority.

- The gap between available methodology and current practice is documented and not attributable to missing tools.

- Failure modes in chronic-condition pharmacology are consequential, not academic.
- The remainder of the paper develops, applies, and evaluates the decomposition framework.

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.

The case we develop in the rest of this paper is therefore that the gap is real, that the methodological tools to fill it are mature, and that the failure modes of the lumped analysis are large enough in chronic-condition pharmacology to make the decomposition a methodological priority rather than an academic refinement.

1.4 The neurobiological and epidemiological reality of the components

[12]

Each component of the decomposition reflects a real causal mechanism documented in independent literatures.

- *BR* corresponds to pharmacological action; *PB* to neurobiologically real belief-driven mechanisms; *TV* to natural-history drift.
- The components are not statistical constructs but distinct causal pathways with separate literatures.
- The section reviews the evidence for each component before establishing the formal model.

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.

The decomposition is not a statistical convenience. Each component reflects a real causal mechanism documented in independent literatures, and we shall briefly recall each in turn before turning to the formal model.

[13]

***BR* is the pharmacological target of regulation; *PB* is neurobiologically real and phase-modulated, which is precisely the design contrast that makes the decomposition identifiable.**

- *PB* is mediated by endogenous opioid/dopamine release, HPA-axis regulation, and top-down attentional shifts.
- Wager et al. neurologic pain signature provides a clean fMRI dissociation between pharmacological and expectancy contributions.
- Phase-dependent belief variation (open-label vs. blinded) is the design contrast that supplies *PB* identifiability.

440 *Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod*
441 *tempor incididunt ut labore et dolore magna aliqua.*

442 The biological response component represents the pharmacological ac-
443 tion of the active compound: the chemical binding, downstream sig-
444 nalling, and physiological response that a regulatory authority cares
445 about when it grants approval. The placebo-belief response reflects
446 neurobiologically real mechanisms documented across pain, depres-
447 sion, Parkinson's, and PTSD research^{38–41}, mediated by belief-driven
448 release of endogenous opioids and dopamine, anticipatory regulation
449 of the hypothalamic-pituitary-adrenal axis, and top-down attentional
450 shifts in symptom perception. Direct neuroimaging evidence is in-
451 structive here:⁴² developed an fMRI-based 'neurologic pain signature'
452 (NPS) that tracks nociceptive input and is reduced by the opioid ag-
453 onist remifentanyl, and the individual-participant meta-analysis of⁴³
454 found that placebo treatments produce moderate reductions in re-
455 ported pain but only negligible reductions in the NPS, providing a
456 neural dissociation between the pharmacological and expectancy con-
457 tributions to a clinical pain response. The placebo response is hetero-
458 geneous across patients with biological correlates^{9,10} and is modulated
459 by trial phase: when belief varies, as it does between open-label and
460 blinded-discontinuation conditions, the placebo response varies, and
461 the modulation is exactly the design contrast that makes the decom-
462 position identifiable.

463 [14]

464 ***TV is the untreated counterfactual trajectory; its heterogene-***
465 ***ity is the central object of precision-medicine inference and***
466 ***must be modelled, not absorbed into noise.***

- 467 • *TV* sign depends on disease class: positive for acute recovery,
468 near-zero for stable chronic, negative for progressive conditions.
- 469 • Selection-driven enrolment at peak severity guarantees positive
470 *TV* in early trial weeks regardless of disease class.
- 471 • The PATH statement identifies treatment-effect heterogeneity
472 modelling as mandatory for precision-medicine inference.

473 *Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod*
474 *tempor incididunt ut labore et dolore magna aliqua.*

475 The time-variant natural-history component reflects whatever trajec-
476 tory the participant would have followed without treatment. For acute
477 conditions this trends positive (recovery on intrinsic timescale); for
478 chronic stable conditions it is near zero on the population mean but
479 with substantial participant-level variance; for chronic progressive con-
480 ditions it trends negative; and for selection-driven enrolment processes
481 it is positive in the early weeks of the trial regardless of underly-
482 ing disease class, because of regression to the mean^{11,13}. The PATH

statement^{14,15} highlights that this heterogeneity is itself the central object of precision-medicine inference: heterogeneity of treatment effect must be modelled, not absorbed into noise, if the precision-medicine question is to be answered.

1.5 What this paper contributes

[15]

The paper makes four contributions spanning formalisation, identifiability mapping, tractable methodology, and quantification of the inferential cost of lumping.

- Sections 2–3 formalise the decomposition; section 4 maps it to design contrasts; section 5 develops tractable analysis methods.
- Sections 6–7 provide worked examples and a Monte Carlo simulation study; section 8 applies the framework to biomarker validation.
- A companion pedagogical document provides extended worked examples for first-time readers.

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.

We make four contributions, each addressed in a section below. We begin in section 2 with the formal model and notation. Section 3 develops each component in turn, with biological motivation, parametric form, and identifiability conditions. Section 4 maps the design-contrast structure of the standard hybrid open-label-plus-blinded-discontinuation design onto the identifiability requirements of the decomposition. Section 5 develops the analysis-side methodology, including the linear-mixed-effects approximations and the phase-by-treatment indicator extension that retains BR-PB sensitivity at low parametric cost. Section 6 presents the worked examples, and section 7 presents the simulation study. Section 8 discusses applications to predictive-biomarker validation. The companion pedagogical document at docs/24-component-decomposition-pedagogy.md provides an extended worked-example treatment for readers approaching the material for the first time.

2 Notation and the formal model

[16]

This section fixes compact notation and states the formal model; a fuller worked-example treatment is in the companion pedagogical document.

- Definitions are precise enough to support the identifiability and analysis-side arguments in later sections.
- The presentation is kept compact; extended derivations are deferred to the companion document.
- The companion pedagogical document provides numerical worked examples at each formal step.

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.

In this section we shall fix notation and state the formal model. The three components are defined precisely enough to make the identifiability and analysis-side arguments in later sections rigorous, but we have endeavored to keep the presentation compact. A fuller treatment, with worked numerical examples at each step, is available in the companion pedagogical document at docs/24-component-decomposition-pedagogy.md.

[17] where Y_{it} is the observed symptom score, BL_i is participant i 's baseline score, the three components BR , PB , TV are non-negative reductions in symptom severity attributable to the three causal mechanisms identified in section 1, and ε_{it} is observation-level noise. The sign convention is that an increase in any component reduces Y ; all three components are zero at $t = 0$ by construction (no exposure yet to drug, no belief activation, no elapsed natural history) and grow over the duration of the trial.

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.

For participant i at trial timepoint t :

$$Y_{it} = BL_i - [BR_{it} + PB_{it} + TV_{it}] + \varepsilon_{it},$$

[18]

Each component is driven by its own time-on-state variable, making the components structurally distinct even when they overlap in calendar time.

- BR_{it} accumulates with time-on-drug; PB_{it} with time-on-belief scaled by expectancy $\eta(\text{phase})$; TV_{it} with calendar time from trial entry.
- The distinct time indices are the structural source of identifiability from phase-contrast data.
- All components are zero at $t = 0$ by construction and grow over the trial duration.

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod

559 *tempor incididunt ut labore et dolore magna aliqua.*

560 Each component is a participant-specific function of its own time-
561 on-state variable. BR_{it} depends on time-on-drug, the cumulative
562 exposure to active medication since drug initiation. PB_{it} depends
563 on time-on-belief, modulated by a phase-specific expectancy factor
564 $\eta(\text{phase})$ that captures the patient's confidence that they are receiv-
565 ing active treatment. TV_{it} depends on time-since-trial-entry, calendar
566 time elapsed regardless of treatment status.

567 [19] rescaled so that $C(0) = 0$ and $C(t) \rightarrow m_C(1 - e^{-d_C})$ as $t \rightarrow \infty$,
568 for $C \in \{BR, PB, TV\}$. The three Gompertz parameters (m_C, d_C, r_C)
569 are participant-specific random effects drawn from population distri-
570 butions, and the participant-level heterogeneity in these distributions
571 is the principal source of between-person response variability that the
572 framework is designed to characterise.

573 *Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod*
574 *tempor incididunt ut labore et dolore magna aliqua.*

575 Each component follows a modified Gompertz curve:

$$C(t) = m_C \exp(-d_C \exp(-r_C t)) - m_C \exp(-d_C),$$

576 [20]

577 **The residual term has AR(1) structure and the three com-**
578 **ponents are permitted to co-vary through cross-component**
579 **covariances.**

- 580 • Within-participant residuals follow $\text{Cov}(\varepsilon_{it}, \varepsilon_{i,t+h}) = \sigma_\varepsilon^2 \rho^{|h|}$.
581 • Within-time and across-time cross-component covariances permit
582 participant-level heterogeneity in one component to co-vary with
583 the others.
584 • This structure accommodates biologically plausible correlations,
585 such as high- BR patients also tending toward high- PB .

586 *Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod*
587 *tempor incididunt ut labore et dolore magna aliqua.*

588 ε_{it} is a within-participant residual term with AR(1) serial-correlation
589 structure $\text{Cov}(\varepsilon_{it}, \varepsilon_{i,t+h}) = \sigma_\varepsilon^2 \rho^{|h|}$. The three components are permit-
590 ted to be jointly correlated through within-time and across-time cross-
591 component covariances c_{cf1t} and c_{cfct} respectively, so that participant-
592 level heterogeneity in any one component can co-vary with heterogene-
593 ity in the others.

594 The principal psychometric and pharmacological terms used below are sum-
595 marised in Table 1.

Table 1: Principal psychometric and pharmacological terms used in the three-component decomposition.

Term	Meaning
BR	Biological response: pharmacological reduction in symptom score attributable to active drug.
PB	Placebo-belief response: reduction attributable to the patient’s belief that they are receiving treatment.
TV	Time-variant component: reduction (or increase) attributable to the natural-history trajectory of the disease.
$\eta(\text{phase})$	Expectancy multiplier for <i>PB</i> : 1 in open-label on-drug, ≈ 0.5 in blinded discontinuation, between on-drug and off-drug.
BL_i	Participant i ’s baseline (pre-treatment) symptom score.
D_{it}	Continuous-decay drug indicator on the analysis side: 1 on-drug, $\exp(-\lambda t_{sd,it})$ off-drug.
$t_{sd,it}$	Time since drug discontinuation (off-drug only).
λ	Carryover decay rate, with half-life $t_{1/2} = \ln 2 / \lambda$.

3 The three components

[21]

This section presents each component with its biological motivation, Gompertz parameterisation, and identifiability conditions to establish the model as causally interpretable.

- The three components are presented in turn: *BR* (biological response), *PB* (placebo-belief), *TV* (natural history).
- Each component is grounded in independent biological literature before its parametric form is stated.
- The section’s purpose is to demonstrate that the model sum reflects causal structure, not mathematical convenience.

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.

In this section we shall present the three components in turn, each with its biological motivation, the modified Gompertz parameterisation, and the design contrasts that identify it. The treatment is organised so that the reader can see, by the end of this section, that the formal model is a sum of three causally interpretable trajectories rather than a mathematical convenience.

3.1 Biological response (BR)

[22]

***BR* is the pharmacological target of regulatory approval and biomarker stratification, independent of belief or natural history.**

- *BR* arises from chemical action on the body and is zero for inert pills regardless of patient expectation.

- It is the estimand that approval decisions, dose selection, and biomarker-guided stratification require.
- Separating BR from PB and TV is the central precision-medicine problem the framework addresses.

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.

BR is the physiological reduction in symptom score attributable to the active medication's chemical action on the body. It is independent of belief, suggestion, and natural history: a participant who somehow received active drug without knowing it would still develop a measurable BR , and a participant who took an inert sugar pill while believing it was active drug would have $BR = 0$ no matter how strong their expectation. Methodologically, BR is the quantity that drug regulators and prescribers want: approval decisions, dose selection, and biomarker-guided patient stratification are all framed around BR .

[23]

PK/PD considerations produce a saturating BR trajectory with a slow initial rise, a peak-growth phase, and a sharp drop to zero on discontinuation.

- Plasma steady state is reached within a few half-lives; pharmacodynamic integration (e.g. receptor adaptation) adds additional lag.
- For prazosin-PTSD, Raskind et al. (2013) calibrate the onset to several weeks of sustained exposure.
- The saturating shape motivates the Gompertz over linear or simple exponential alternatives.

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.

The qualitative shape of the BR trajectory follows from pharmacokinetic and pharmacodynamic considerations. Drug concentration in plasma reaches steady state within a small multiple of the elimination half-life; brain concentration follows plasma with a short delay; the downstream pharmacodynamic response (reduction in symptom-relevant neural activity, in the case of prazosin's effect on noradrenergic tone during REM sleep) typically requires several days to weeks of sustained exposure to integrate fully⁴⁴. The result is a saturating curve: slow rise in the first week, steepest growth in the second to third week, plateau as receptor adaptation completes, and a decay back toward zero on discontinuation governed by the carryover half-life $t_{1/2} = \ln 2/\lambda$ rather than an instantaneous drop. The rate of this off-drug decay is the object of the carryover machinery used throughout this project; only in the limit of a short half-life relative to the

measurement spacing does it approximate the sharp step assumed in the worked example of section 6.

[24]

The three Gompertz parameters (m_{BR}, r_{BR}, d_{BR}) capture ceiling, onset rate, and climb shape respectively, calibrated to prazosin-PTSD reference values.

- m_{BR} is the biological ceiling (maximum reduction at infinite exposure); r_{BR} the onset rate; d_{BR} the displacement shaping the early climb.
- Prazosin-PTSD calibration yields $m_{BR} \approx 11$, $r_{BR} \approx 0.42/\text{week}$, $d_{BR} \approx 5$.
- These are participant-specific random effects whose population distributions encode between-person response heterogeneity.

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.

The modified Gompertz form $BR(t) = m_{BR} \exp(-d_{BR} \exp(-r_{BR}t)) - m_{BR} \exp(-d_{BR})$ captures this profile compactly. The asymptote $m_{BR}(1 - e^{-d_{BR}})$ is the biological ceiling, the largest reduction the drug can produce at infinite exposure; the scale parameter m_{BR} equals that ceiling up to the factor $(1 - e^{-d_{BR}})$, which for the calibrated $d_{BR} \approx 5$ exceeds 0.99. The onset rate r_{BR} controls how rapidly the participant approaches the ceiling: high r_{BR} means a fast onset (effect plateaus in one to two weeks), low r_{BR} a slow climb (six to eight weeks). The displacement d_{BR} shapes the climb: high d_{BR} gives an early-onset curve that rises sharply at first, low d_{BR} a more gradual climb. For PTSD prazosin, calibrated values from the³⁴ reference dataset are $m_{BR} \approx 11$ (eleven points of nightmare-score reduction at saturation), $r_{BR} \approx 0.42$ per week (half-maximum at roughly four to five weeks of sustained exposure), and $d_{BR} \approx 5$ (moderately steep early climb).

[25]

The Gompertz is preferred over linear and simple-exponential alternatives because it captures the asymmetric slow-start saturation profile of chronic-condition pharmacodynamics.

- Linear $BR = \beta t$ is biologically implausible past the first few weeks due to unbounded growth.
- Exponential $BR = m(1 - e^{-rt})$ lacks the slow-start phase from receptor adaptation and gene-expression changes.
- The Gompertz interpolates between both limits and its three-parameter form avoids overfitting; sensitivity analysis is reported separately in 07-gompertz-evaluation.

705 *Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod*
706 *tempor incididunt ut labore et dolore magna aliqua.*

707 Two alternative families that do not capture the asymmetric saturation
708 profile are worth noting for contrast. A linear function $BR = \beta t$
709 climbs without bound and is biologically implausible past a few weeks.
710 An exponential decay $BR = m_{BR}(1 - \exp(-rt))$ is acceptable for
711 fast-onset drugs but lacks the slow-start phase that characterises most
712 chronic-condition responses, where receptor adaptation, feedback regulation,
713 and gene-expression changes produce a measurable lag. The Gompertz
714 interpolates between exponential decay (high d limit) and linear-in- t growth
715 (low d limit), and its three-parameter form is flexible enough to fit a wide
716 range of pharmacodynamic profiles without overfitting. The Gompertz is a
717 deliberate choice over the two parametric forms most familiar from
718 pharmacokinetic-pharmacodynamic modelling. The sigmoid $E_{\max}$ (Hill) model⁴⁵
719 describes the dependence of effect on concentration (a dose-response curve),
720 whereas $BR(t)$ here is a time-course of accumulating effect at fixed dosing;
721 and the indirect-response (turnover) models that do describe effect time-
722 courses are specified through differential equations on a hidden response
723 variable rather than in closed form. The modified Gompertz retains the
724 slow-start, saturating shape those models produce while remaining a closed-
725 form, three-parameter mean function that is tractable inside a mixed-effects
726 likelihood. Doc 25 of the project documentation provides the full historical
727 and biological context for the family choice and quantifies the sensitivity
728 of pmsimstats inference to alternative families; that work is reported
729 separately as 07-gompertz-evaluation. The headline result of that evaluation
730 is reassuring for the present framework: in a 1{,}000-replicate-per-cell
731 simulation that generated data under the logistic, hyperbolic-tangent, and
732 piecewise-linear breakpoint families and analysed them under the standard
733 linear-mixed model, the trajectory family was essentially irrelevant to the
734 power of the biomarker-treatment interaction test (the family-to-family
735 spread fell within two Monte Carlo standard errors of sampling noise) and
736 to type I error (0.025 to 0.036 across null cells, with no family-specific
737 inflation). The conclusions drawn here are therefore not artefacts of the
738 Gompertz form, though the regimes the family cannot represent (cyclical
739 or biphasic trajectories) remain genuine exclusions.

742 [26]

743 ***BR* is identified by the blinded-discontinuation contrast,**
744 **where drug removal collapses *BR* to zero while *PB* remains**
745 **intact.**

- 746 • Within-participant on-drug versus off-drug contrasts are the pri-
747 mary identification mechanism.

- The blinded-discontinuation phase provides the cleanest contrast: BR drops, PB is preserved.
- This contrast does not require the analyst to model PB or TV explicitly to recover BR .

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.

The BR component is identified by within-participant on-drug versus off-drug contrasts, formalised in section 4 below. The most direct contrast is the blinded-discontinuation phase: when a participant is silently switched to placebo while still believing they are on active drug, BR decays toward zero as the drug clears while PB remains roughly intact, and the difference between on-drug and off-drug response in that window is a relatively clean estimate of BR that does not require the analyst to model PB or TV explicitly. The decay is not instantaneous: with the carryover half-life $t_{1/2} = \ln 2 / \lambda$ of the analysis-side decay term, residual BR persists into the early off-drug observations, so the contrast is clean only once the discontinuation window is long relative to $t_{1/2}$, and the drug state is best carried by the continuous-decay indicator D_{it} of Table 1 rather than a hard on/off indicator.

3.2 Placebo-belief response (PB)

[27]

PB is the strict placebo response: improvement that persists when the active drug is silently replaced, provided the patient's belief is unchanged.

- PB is belief-dependent, not drug-dependent; it would survive a silent switch to placebo under sustained belief.
- It is distinct from BR in causal origin: BR requires chemical action; PB requires only belief.
- The definition is strict and operational, not vague or colloquial.

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.

PB is the reduction in symptom score that arises from the patient's belief that they are receiving active treatment, independent of what they are actually receiving. That is, it is the placebo response in its strict sense: the part of the improvement that would persist if the active drug were silently replaced with an inert pill, provided the patient continued to believe they were on the active drug.

[28]

PB reflects neurobiologically real mechanisms triggered by

belief, not pharmacology; neuroimaging and genetic evidence confirm it is a structured biological phenomenon.

- Belief-driven endogenous opioid/dopamine release, HPA-axis modulation, and top-down attentional shifts are all documented mechanisms.
- Wager et al. neurologic pain signature dissociates pharmacological from expectancy contributions via fMRI.
- COMT genotype predicts placebo responsiveness, confirming biological structure rather than measurement artefact.

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.

It is worth being precise about what *PB* is and is not. *PB* is not ‘fake’ improvement; it is a phenomenon clinicians knowingly exploit, a national survey of US internists and rheumatologists finding roughly half report regular use of placebo treatments⁴⁶. As the introduction sets out, the placebo response is mediated by well-characterised neurobiological pathways^{38–41,47}: belief-driven release of endogenous opioids and dopamine, anticipatory hypothalamic-pituitary-adrenal regulation, and top-down attentional shifts, dissociated from pharmacology by the⁴² neurologic pain signature⁴³ and shown to have genetic correlates^{9,10}. What makes these mechanisms ‘placebo’ rather than ‘drug’ is that they are triggered by belief rather than by pharmacology.

[29]

***PB* is identifiable from *BR* only when belief and treatment allocation differ, as in the blinded-discontinuation phase.**

- The blinded-discontinuation phase is the canonical contrast: patients cannot tell whether they remain on active drug.
- Belief drops to a partial level ($\eta \approx 0.5$) because the patient’s expectation is hedged, not absent.
- Designs without a phase that decouples belief from treatment cannot separately identify *PB* from *BR*.

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.

For trial design, the relevant point is that *PB* is separately identifiable from *BR* if and only if the trial includes a phase in which the patient’s belief about treatment differs from the actual treatment they are receiving. The blinded-discontinuation phase is the canonical such phase: the patient is told that, at some unannounced point during the window, they may be silently switched to placebo, and they cannot reliably tell whether they are still receiving active drug. Belief in

829 treatment correspondingly drops to a partial level (multiplier roughly
830 0.5) because the patient's expectation is hedged.

831 [30]

832 **The $\eta = 0.5$ blinded-phase expectancy is a modelling assumption**
833 **operationalising response expectancy, not a measured**
834 **quantity, and requires sensitivity analysis.**

- 835 • The construct is response expectancy in the sense of Kirsch
836 (1985); to our knowledge the residual expectancy under blinded
837 discontinuation has not been measured directly.
- 838 • Open-label placebo trials confirm clinically meaningful *PB*-only
839 responses even under explicit disclosure, bounding the residual
840 response away from zero.
- 841 • Sensitivity analyses varying η are mandatory before clinical con-
842 clusions depend on this single parameter.

843 *Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod*
844 *tempor incididunt ut labore et dolore magna aliqua.*

845 *PB* also follows a Gompertz curve, scaled by the expectancy factor:

$$PB(t \mid \text{phase}) = \eta(\text{phase}) \cdot m_{PB} \exp(-d_{PB} \exp(-r_{PB}t)) + (\text{offset to start at zero}).$$

846 Typical expectancy values are $\eta = 1.0$ in open-label on-drug, $\eta \approx 0.5$ in
847 blinded discontinuation, and η between 0 and 0.5 in open-label placebo
848 (the residual expectation a patient retains when told the previous ac-
849 tive phase was beneficial). The quantity η operationalises response
850 expectancy in the sense of⁴⁸, the patient's anticipation of benefit that
851 the expectancy literature treats as a determinant of experienced symp-
852 tom change. The choice of $\eta = 0.5$ for the blinded phase is a modelling
853 assumption rather than a calibrated quantity: to our knowledge the
854 residual expectancy a patient carries when uncertain about allocation
855 has not been measured directly as a fraction of the open-label level,
856 and we adopt 0.5 as a deliberately central placeholder. The avail-
857 able indirect evidence is nonetheless that the residual response is sub-
858 stantial. Open-label placebo trials^{35–37} demonstrate that a clinically
859 meaningful response persists even when patients are explicitly told
860 they are receiving an inert intervention, which bounds the residual
861 belief-driven response away from zero. Because the value of η is not
862 empirically pinned, sensitivity analyses varying it across its plausible
863 range are mandatory before any conclusion is allowed to depend on
864 this single number.

865 [31]

866 **Kaptchuk et al. (2008) provide the structural precedent; the**

867 **BR-PB-TV framework extends the same decomposition logic**
868 **to the full treatment response.**

- 869 • Kaptchuk decomposed the placebo response itself into observa-
870 tion, ritual, and relationship components empirically.
- 871 • The present framework applies analogous decomposition logic at
872 the level of drug, placebo, and natural history.
- 873 • Both frameworks reject the 'expectancy lump' model in favour of
874 identifiable additive parts.

875 *Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod*
876 *tempor incididunt ut labore et dolore magna aliqua.*

877 The decomposition framework as proposed here has a structural ana-
878 logue in the placebo-research literature.⁵ decomposed the placebo re-
879 sponse itself into observation, ritual, and patient-practitioner relation-
880 ship components in a randomised trial of acupuncture-style placebo for
881 irritable bowel syndrome, demonstrating empirically that the placebo
882 response is not a single 'expectancy' lump but a sum of identifiable
883 parts. The present BR-PB-TV framework is analogous in spirit but
884 operates at the level of the full treatment response (drug, placebo,
885 natural history) rather than within the placebo response alone.

886 [32]

887 ***PB* variance scales with η , not just its mean; ignoring this**
888 **heteroscedastic structure miscalibrates standard errors and**
889 **downstream biomarker tests.**

- 890 • Phase-dependent variance follows $SD(PB \mid \text{phase}) = \eta(\text{phase}) \cdot$
891 $\sigma_{PB}^{\text{baseline}}$.
- 892 • Open-label phases have high between-patient *PB* variance;
893 blinded phases have low variance.
- 894 • Ignoring heteroscedasticity deflates standard errors in open-label
895 and inflates them in blinded phases, with consequences for CI
896 coverage and biomarker test calibration.

897 *Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod*
898 *tempor incididunt ut labore et dolore magna aliqua.*

899 A subtle property of *PB* is that its variance also scales with η ,
900 not just its mean. When η is high, there is substantial variability
901 across patients in how strongly the placebo response is expressed;
902 high-suggestibility patients show large *PB*, low-suggestibility pa-
903 tients show essentially none. When η is low (blinded), the placebo
904 response is uniformly attenuated and the patient-to-patient vari-
905 ance shrinks accordingly. The model therefore parameterises
906 $SD(PB \mid \text{phase}) = \eta(\text{phase}) \cdot \sigma_{PB}^{\text{baseline}}$. Ignoring this heteroscedastic
907 structure deflates standard errors in the open-label phase and inflates
908 them in the blinded phase, with downstream consequences for

confidence-interval coverage and for the calibration of any biomarker test built on top of the analysis.

3.2.1 Additivity, subadditivity, and the BR-PB interaction

[33]

Additivity is a modelling assumption, not biological fact; the section examines whether a $BR \times PB$ interaction term is warranted.

- The formal model writes total response as a sum; this section asks whether the BR - PB interaction is zero.
- The question has inferential consequences for biomarker validation and for what the phase-augmented analysis captures.
- The treatment is preparatory for the generalised interaction model introduced later in this section.

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.

In this section we shall broach a question that the formal model leaves open: whether the three components contribute additively to symptom reduction, or whether there is a meaningful interaction between BR and PB that the additive specification misses.

[34]

The additive model may be wrong: published literature documents subadditivity between pharmacological and placebo-belief effects in some clinical contexts.

- Subadditivity means the combined $BR + PB$ effect is less than the sum of each component measured separately.
- Evidence from Kaptchuk et al. and Benedetti et al. shows the phenomenon is context-dependent, not universal.
- Formal test: $\partial Y / \partial (BR \cdot PB) > 0$ in a saturated mean-structure model.

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.

The decomposition as written in section 2 treats the three components as additive contributors to symptom reduction:

$$Y_{it} = BL_i - (BR_{it} + PB_{it} + TV_{it}) + \varepsilon_{it}.$$

The additivity is a modelling assumption, not a fact about biology, and it is worth stating that the assumption may be wrong in directions the framework can absorb and in directions it cannot. Additiv-

ity is also scale-dependent in the sense emphasised by⁴⁹: components that combine non-additively on the raw symptom scale may combine additively on a transformed (for example logarithmic) scale, so an apparent $BR \times PB$ interaction can be partly an artefact of the chosen response metric rather than a biological fact. That belief and pharmacology interact rather than act in isolation is demonstrated directly by⁵⁰, who showed with neuroimaging that positive treatment expectation roughly doubled the analgesic benefit of the opioid remifentanyl while negative expectation abolished it, so the two channels are not in general separable by simple addition. More specifically, the published literature on placebo-drug interactions has accumulated evidence that the combined effect of receiving an active drug and believing one is receiving an active drug is, in some clinical contexts, less than the sum of the two components measured separately^{5,38}. The phenomenon is typically termed subadditivity: $\partial Y / \partial (BR \cdot PB) > 0$ when both components are present and their interaction term is included in a saturated mean-structure model.

[35]

Three candidate mechanisms for subadditivity are shared neural substrate, scale ceiling effects, and belief-dynamics adaptation.

- Shared pain-modulation circuits produce saturating occupancy when both BR and PB activate them simultaneously.
- Scale ceiling effects prevent registration of additional symptom reduction once $BR + PB$ saturates the measurement range.
- Belief-dynamics adaptation: unexpectedly large BR can recalibrate patient expectation downward, attenuating ongoing PB .

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.

Three mechanisms have been proposed in the placebo-research literature to explain subadditivity. First, a shared neural substrate: if the descending pain-modulation circuits that mediate placebo analgesia overlap with the circuits modulated by the active drug, simultaneous activation by both mechanisms produces less marginal benefit than activation by either alone, in a saturating-occupancy sense. Second, ceiling effects on the symptom scale: a clinical scale that caps at zero (no symptoms) cannot register additional reduction once a participant has reached that cap, so any participant whose $BR + PB$ already saturates the scale will register no further reduction even if BR or PB would, in isolation, predict more. Third, belief-dynamics adaptation: the patient's expectation may adapt downward when the active drug produces a strong response that exceeds the expected effect, reducing the ongoing PB contribution; equivalently, unexpectedly large BR

988 may attenuate PB via expectation recalibration.

989 [36] where $\gamma < 0$ encodes subadditivity, $\gamma = 0$ recovers the additive
990 form, and $\gamma > 0$ encodes superadditivity, is the natural extension.
991 The interaction coefficient γ is identifiable from the same trial-design
992 contrasts that identify the marginal components: open-label phases
993 supply $BR \cdot PB$ products at high η , blinded phases supply $BR \cdot PB$
994 products at attenuated η , and the γ coefficient is estimated from the
995 difference between the two regimes once the marginal components are
996 accounted for.

997 *Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod*
998 *tempor incididunt ut labore et dolore magna aliqua.*

999 Whichever mechanism dominates in a given clinical context, the subad-
1000 ditivity question is empirical and design-dependent. Within the BR-
1001 PB-TV framework, subadditivity manifests as a non-zero coefficient
1002 on a $BR \times PB$ interaction term in the conditional-mean specification.
1003 The additive form treats this interaction as zero by construction; a
1004 generalised form

$$Y_{it} = BL_i - [BR_{it} + PB_{it} + TV_{it} + \gamma \cdot BR_{it} \cdot PB_{it}] + \varepsilon_{it},$$

1005 [37]

1006 **Subadditivity biases biomarker predictive accuracy estimates**
1007 **and is partly absorbed by the phase-augmented analysis even**
1008 **without explicit γ estimation.**

- 1009 • Under subadditivity, the biomarker-by- BR association is partly
1010 cancelled, inflating apparent predictive accuracy in the additive
1011 model.
- 1012 • Phase designs attenuating PB (blinded, crossover) push the effec-
1013 tive biomarker-treatment estimate closer to the true biomarker-
1014 by- BR value.
- 1015 • The phase-augmented analysis in section 5 absorbs subadditivity
1016 as a side benefit.

1017 *Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod*
1018 *tempor incididunt ut labore et dolore magna aliqua.*

1019 For trial design, two consequences follow. First, a biomarker that
1020 predicts BR in an additive-decomposition analysis may overstate or
1021 understate predictive accuracy if the true mechanism is subadditive:
1022 under subadditivity, the biomarker-driven increase in BR is partly
1023 cancelled by the $\gamma \cdot BR \cdot PB$ interaction, and the marginal biomarker-
1024 treatment association is smaller than the biomarker-by- BR associa-
1025 tion. Second, trial designs that attenuate PB in some phases (blinded

discontinuation, crossover) implicitly down-weight the subadditivity contribution to the response in those phases, which means that the effective biomarker-treatment association estimated from those phases is closer to the biomarker-by- BR association than is the open-label estimate. The phase-augmented analysis recommended in section 5 below therefore has the side benefit of partly absorbing subadditivity, even when γ is not estimated explicitly.

[38]

A forthcoming simulation study will quantify how bias in the biomarker-treatment interaction scales with γ ; the pre-registration predicts linear scaling.

- Bias in $\hat{\beta}_{bm:D}$ is expected to scale linearly with γ at small $|\gamma|$.
- Under subadditivity, bias sign is opposite to the true biomarker-by- BR effect; under superadditivity it is aligned.
- Empirical quantification is deferred to the production simulation programme in section 7.

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.

A simulation study quantifying the magnitude of bias in the biomarker-treatment interaction estimate when the additive analysis is fitted to data generated under non-zero γ is forthcoming as part of the simulation programme described in section 7. The pre-registration is that bias scales linearly with γ at small $|\gamma|$, with the sign opposite to the true biomarker-by- BR effect under subadditivity and aligned with it under superadditivity.

[39]

Non-zero $\hat{\gamma}$ does not establish true subadditive interaction; latent-class structure with class-correlated components produces the same marginal pattern.

- If class membership correlates with both BR and PB strength, the marginal covariance of $BR \cdot PB$ mimics a direct interaction term.
- This confounding is invisible from the lumped-analysis perspective but is identifiable in a class-aware analysis.
- Apparent subadditivity can arise from latent population heterogeneity with no direct BR - PB interaction in the class-conditional DGP.

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.

A subtle but inferentially important point merits attention here: a non-zero $\hat{\gamma}$ recovered from an additive-extension analysis does not,

on its own, establish that the underlying mechanism is subadditive interaction in the strict sense. The same marginal pattern can arise from latent population structure that the analyst is not modelling. If the trial population is a mixture of treatment-responders and non-responders (in the latent-class sense developed at length in [analysis/report/03-latent-class-mixture-application/](#)), and class membership correlates with both BR strength and PB strength (responder-class participants have both higher BR and higher PB on average), then the marginal expected value of $BR \cdot PB$ over the unobserved class label has a non-zero covariance term whose magnitude depends on the gating function $\pi(B)$. From the lumped-analysis perspective this looks exactly like a γ -weighted $BR \times PB$ interaction even though no such direct interaction was specified in the class-conditional DGP. That is, the latent-class structure is one mechanism that produces apparent subadditivity in a one-component analysis, regardless of whether the underlying biology has a true direct BR - PB interaction.

[40]

Distinguishing true subadditivity from latent-class confounding requires either a class-aware analysis or the phase contrasts the BR-PB-TV framework provides.

- The two mechanisms produce qualitatively similar marginal effects but are inferentially distinct.
- $\hat{\gamma}$ should be reported alongside a class-aware sensitivity analysis, not as standalone evidence for mechanistic subadditivity.
- The latent-class paper’s `pb_class_correlation` simulation axis is the direct test; cross-paper synthesis is the cleanest disentanglement.

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.

The two mechanisms, namely true subadditive interaction and latent-class structure with class-correlated components, produce qualitatively similar marginal effects but are inferentially distinct. Distinguishing them requires either a class-aware analysis that explicitly models the latent population structure (as the latent-class paper develops at length), or a phase contrast that separates the mechanisms by exploiting the differential expectancy modulation of PB that the BR-PB-TV framework provides. Under the additive-extension analysis fitted to mixture-DGP data with class-correlated components, $\hat{\gamma}$ is identified up to confounding with the class structure, and the analyst should report $\hat{\gamma}$ alongside a class-aware sensitivity analysis rather than as standalone evidence for mechanistic subadditivity. The Study 1 simulation in the latent-class paper includes a `pb_class_correlation` axis that varies

$\text{Cor}(BR, PB)$ across class labels; that axis is the direct simulation-side test of the latent-class generation of apparent subadditivity, and the cross-paper synthesis between the two papers is the cleanest way to disentangle the two mechanisms.

3.3 Time-variant natural-history component (TV)

[41]

TV is the untreated counterfactual trajectory; it is participant-specific, structured, and can be positive, negative, or near-zero depending on disease class.

- TV is the symptom change that would have occurred without treatment, integrated over the trial duration.
- It varies in sign across disease classes and across patients within a class.
- It is independent of treatment status and belief, distinguishing it structurally from PB .

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.

TV is the reduction in symptom score that would have happened even if the patient had received no treatment at all. It is the participant's natural-history trajectory, integrated over the duration of the trial. For some participants TV is positive (symptoms naturally improve over the trial because of life events, therapy, time-in-condition, or regression to the mean); for others TV is negative (symptoms naturally worsen because of trauma anniversaries, accumulating stress, or progressive underlying disease).

[42]

TV is a structured participant-specific signal, not noise; its distinguishing feature from PB is independence from treatment status and belief.

- Unlike ε_{it} , TV is participant-level (slow trajectory) and requires explicit modelling, not absorption into noise.
- Off-drug baselines and follow-up windows are the primary data sources for estimating TV .
- TV is modulated by neither treatment assignment nor patient belief, whereas PB is modulated by both.

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.

Like PB , TV is not noise. It is a structured participant-specific signal that the model can in principle estimate from data, given sufficient repeated measurements and a trial design that includes a long enough

1150 off-drug baseline or follow-up window. What distinguishes TV from
1151 PB is its origin: TV is independent of treatment status and of belief
1152 about treatment, whereas PB is modulated by both.

1153 3.3.1 When does natural history actually exist?

1154 [43]

1155 **The shape of TV depends on disease class, trial timescale,**
1156 **and selection process; none of these can be assumed and all**
1157 **must be recovered empirically.**

- 1158 • The assumption that subjects improve over time is true for some
1159 conditions and systematically wrong for others.
- 1160 • Disease class, trial timescale, and enrolment selection jointly de-
1161 termine the sign and magnitude of TV .
- 1162 • Recovering the empirical shape of TV is one of the decomposi-
1163 tion's primary objectives.

1164 *Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod*
1165 *tempor incididunt ut labore et dolore magna aliqua.*

1166 A common simplifying assumption in trial analysis is that subjects will
1167 improve over time, with or without treatment. This is true for some
1168 conditions and problematic for others. The shape of TV depends
1169 on the underlying disease class, the trial timescale, and the selection
1170 process by which participants entered the trial. None of these can be
1171 assumed without thought, and the empirical shape of TV is one of the
1172 things the decomposition is designed to recover.

1173 [44]

1174 **TV sign varies systematically by disease class: positive for**
1175 **acute recovery, near-zero for stable chronic, negative for pro-**
1176 **gressive, and structured for cyclical conditions.**

- 1177 • Acute conditions (post-operative pain, acute back pain) resolve
1178 intrinsically; substantial TV -improvement appears in the placebo
1179 arm.
- 1180 • Progressive conditions (Alzheimer's, ALS) require explicit
1181 negative-trajectory modelling; apparent treatment effect is
1182 slowing of progression.
- 1183 • Cyclical/triggered conditions (PTSD with trauma anniversaries,
1184 relapsing-remitting MS) require covariates beyond the Gompertz
1185 form.

1186 *Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod*
1187 *tempor incididunt ut labore et dolore magna aliqua.*

1188 Acute conditions tend toward TV -positive: a sprained ankle, a post-
1189 operative pain syndrome, an episode of acute back pain all resolve on a

timescale of days to weeks even without intervention, and a trial of an analgesic in such a setting will see substantial TV -improvement in the placebo arm. Chronic stable conditions (treated hypertension, controlled chronic asthma, long-standing PTSD without active triggers) tend toward TV -zero on the population mean but with substantial individual variance. Chronic progressive conditions (Alzheimer's disease, ALS, late-stage congestive heart failure, many cancers) tend toward TV -negative: a trial of a neuroprotective agent in early Alzheimer's must explicitly model the negative natural-history trajectory, since the apparent treatment effect is the slowing of progression, not absolute improvement. Cyclical and triggered conditions (PTSD with trauma anniversaries, seasonal-pattern major depression, relapsing-remitting multiple sclerosis) show structured TV that is neither monotone nor stationary, and may require explicit covariates beyond the Gompertz form.

[45]

Selection-induced TV is real and rarely zero even for stable chronic conditions, because enrolment coincides with peak severity triggering help-seeking.

- Regression to the mean alone contributes several points of apparent improvement in early trial weeks.
- Hrobjartsson et al. systematic reviews across 200+ trials show much of conventional placebo response is indistinguishable from natural-history drift.
- Kirsch (2008) and Locher (2015) demonstrate the clinical consequences in antidepressant and late-life depression trials.

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.

Selection-induced TV is real and rarely zero, even when the underlying condition is stable. Patients enrol at moments of clinical engagement that often coincide with recent worsening, so regression to the mean alone produces several points of apparent TV -improvement in the early weeks of a trial. As the introduction documents, the systematic-review evidence of¹¹, ¹², and¹³ (the last covering 202 trials across 60 conditions) and the antidepressant analyses of⁶ and⁷ show that much of what is conventionally attributed to placebo response is indistinguishable from regression to the mean and natural-history drift.

[46]

Trial participation itself induces TV -improvement through Hawthorne, structured-visit, and diary-completion effects, independent of pharmacology or placebo.

- Hawthorne, structured-visit, and diary-completion effects each

1232 contribute symptom improvements attributable to participation,
1233 not treatment.

- 1234 • These contributions are absorbed into TV in the decomposition,
1235 not into BR or PB .
- 1236 • Ignoring these participation-induced effects inflates the apparent
1237 pharmacological and placebo components.

1238 *Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod*
1239 *tempor incididunt ut labore et dolore magna aliqua.*

1240 Trial participation itself also induces TV : the Hawthorne effect, the
1241 structured-visit effect, and the diary-completion effect appear in the
1242 data as TV -improvement that is independent of any pharmacological
1243 or placebo response.

1244 3.3.2 Can we assume no TV effect?

1245 [47]

1246 **Assuming $TV = 0$ is rarely safe in chronic psychiatric or pain**
1247 **trials; the conservative default is to estimate TV from the**
1248 **data.**

- 1249 • Zero- TV assumption is valid only when all four conditions hold:
1250 stable disease, short trial, non-severity-peaked recruitment, no
1251 ancillary care.
- 1252 • For most chronic conditions at least one condition fails; the as-
1253 sumption should be tested, not asserted.
- 1254 • Statistical efficiency lost by estimating TV is smaller than the
1255 bias introduced by assuming it zero.

1256 *Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod*
1257 *tempor incididunt ut labore et dolore magna aliqua.*

1258 But how safe is it to assume TV is zero? The short answer is: rarely
1259 safe, and never safe without empirical support. The longer answer is
1260 that TV may be assumed zero only if the disease is in a stable chronic
1261 phase, the trial is short relative to the disease timescale, the recruit-
1262 ment process does not favour participants at peak severity, and the
1263 trial protocol does not introduce ancillary care that itself improves
1264 symptoms. For most chronic trials in psychiatric or pain conditions,
1265 at least one of these conditions fails. The conservative default is to es-
1266 timate TV from the data, even at the cost of some statistical efficiency,
1267 rather than assume it is zero.

1268 [48]

1269 **The decomposition makes the zero- TV assumption testable**
1270 **and shows that lumping TV into residuals produces auto-**
1271 **correlated, heteroscedastic residuals that bias BR and PB**

standard errors.

- $\hat{m}_{TV}$ with a variance estimate from the model provides an empirical test of the zero- TV assumption.
- Participant-level TV is structured and correlated with the on/off-drug schedule; it is not independent noise.
- Lumping TV into ε produces autocorrelated within-participant residuals and heteroscedastic cross-participant residuals.

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.

The decomposition makes this assumption testable. A model that includes a TV component will return $\hat{m}_{TV}$ with a variance estimate; if both are small relative to $\hat{m}_{BR}$ and $\hat{m}_{PB}$, the analyst has empirical support for the simplifying assumption. If $\hat{m}_{TV}$ is meaningfully large, the assumption was wrong, and the trial needs the TV component to produce unbiased inference about the others. Setting TV to zero and lumping it into ε is worse than estimating it: the participant-level TV is structured (each participant has their own slow trajectory) and not independent of the on-drug-versus-off-drug schedule, so lumping it into noise produces residuals that are autocorrelated within participants and heteroscedastic across them. The resulting standard errors on the BR and PB estimates are wrong in both directions: too small for participants whose TV is small, too large for participants whose TV is large. Estimating TV explicitly fixes both problems simultaneously.

3.3.3 Why a Gompertz, not a linear time covariate?

[49]

A linear week covariate is insufficient for TV because it enforces linearity, ignores participant heterogeneity, and cannot distinguish TV from BR without the explicit decomposition.

- A **week** covariate constrains each additional week to contribute a fixed natural-history change, misrepresenting curved or saturating disease trajectories.
- A single coefficient estimates a population-mean trajectory, ignoring participant-level heterogeneity in natural history.
- Without the decomposition and design contrasts, **week** absorbs both BR and TV , making them inseparable.

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.

A natural objection is that a model with **week** as a covariate already controls for time. But how adequate is such a control? Three reasons

suggest it is not enough. First, a `week` covariate enforces a linear trajectory, constraining each additional week to contribute a fixed amount of natural-history change; real disease trajectories curve, plateau, or accelerate, and the Gompertz form captures the S-shaped or saturating patterns linear time cannot. Second, a single `week` coefficient estimates a population-mean trajectory that ignores the participant-level heterogeneity in natural history. Third, *BR* and *TV* both evolve smoothly over the same trial timescale, and a `week` covariate cannot tell them apart from observational data alone; the design contrasts discussed in section 4 are what supply identifiability, and without the explicit decomposition the `week` coefficient absorbs both.

4 Identifiability and trial-design contrasts

[50]

Identifiability is supplied by trial design; each phase of the hybrid design contributes a different combination of components to the observed trajectory.

- The decomposition is mathematically specified but requires trial-design contrasts to be statistically identified.
- This section characterises the hybrid open-label-plus-blinded-discontinuation design as the primary identifiability vehicle.
- Each of the three phases supplies a distinct mixture of *BR*, *PB*, and *TV*.

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.

The mathematical decomposition above is meaningless unless the data contain enough information to pin down each component separately. Trial design is what supplies that information. In this section we shall characterise the hybrid open-label-plus-blinded-discontinuation design, used as the worked example throughout this paper, and show how each of its three phases contributes a different combination of components to the observed trajectory.

Phase	Drug?	Belief?	Components present
Open-label on drug	Yes	Full	$BR + PB(\eta = 1) + TV$
Blinded discontinuation	Hidden	Hedged	$BR \cdot \mathbf{1}_{\text{on drug}} + PB(\eta = 0.5) + TV$
Open-label crossover	Either	Knows	$BR \cdot \mathbf{1}_{\text{on drug}} + PB(\eta = 1) + TV$

Three phase-by-allocation contrasts isolate PB , BR , and TV respectively; the LME model combines them but the inference is driven by design.

- Open-label versus blinded contrast yields approximately $0.5 \cdot PB$, isolating the belief component conditional on a known blinded-phase η .
- Blinded on-drug versus blinded placebo yields BR once drug carryover has elapsed (since PB and TV match across arms within the blinded phase).
- Off-drug phases yield TV plus residual off-drug PB , with no active BR contribution beyond carryover.

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.

Three phase-by-allocation contrasts then identify the three components. The open-label versus blinded contrast (within drug) yields $PB(\eta = 1) - PB(\eta = 0.5) \approx 0.5 \cdot PB$, which isolates the belief component, but only conditional on a known blinded-phase expectancy η : from a two-phase design just the product $\eta \cdot m_{PB}$ is identified, and separating η from m_{PB} requires the open-label placebo phase. The blinded on-drug versus blinded placebo contrast yields BR once drug carryover has elapsed, since PB and TV match across allocation arms within the blinded phase. All phases off drug yield TV plus any residual off-drug PB , with no active BR contribution beyond carryover and PB at the off-drug expectation level. The $\mathbf{1}_{\text{on drug}}$ indicator in the phase table above is shorthand for the continuous-decay drug state D_{it} defined in Table 1.

This conditioning on η is the framework's principal identification limitation and deserves emphasis rather than a parenthetical. With a two-phase (open-label and blinded) design the data identify only the product $\eta \cdot m_{PB}$, so PB is recovered up to the assumed blinded-phase expectancy; the working value $\eta \approx 0.5$ is a modelling assumption, not an estimate, and absent a belief or expectancy measurement it is not falsifiable from the outcome data alone. The identified set for m_{PB} is accordingly the interval $\{\eta \widehat{m_{PB}} / \eta : \eta \in (0, 1]\}$, which collapses to a point only when η is pinned down. Two design features close the gap: an open-label placebo phase, which supplies a second equation separating η from m_{PB} , and direct measurement of belief, as in the open-hidden and balanced-placebo designs^{51,52} that manipulate or record the patient's treatment expectation. We therefore recommend that any trial relying on the decomposition for a PB -versus- BR attribution either include such a phase or collect an expectancy measure, and that analyses report sensitivity of the attribution to η across its

plausible range.

[52]

Designs lacking specific phases lose identifiability of the corresponding component; the decomposition machinery is only as useful as the design allows.

- A pure open-label design cannot separate BR from PB because both are present at full strength throughout.
- A pure blinded design cannot estimate $\eta = 1$ PB because no phase achieves full-strength belief.
- Trial designers should verify each component has at least one phase in which it is uniquely present before committing to the design.

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.

A trial that lacks any of these phases loses the corresponding identifiability. A pure open-label design (no blinded discontinuation) cannot separate BR from PB , because both are present at full strength throughout. A pure blinded design (no open-label phase) cannot estimate $\eta = 1$ PB , because no phase has belief at full strength. The decomposition machinery is useful only to the extent that the design supplies identifiability, and trial designers should evaluate any proposed design by checking that each component has at least one phase or contrast in which it is uniquely present.

[53]

Design-contrast identifiability is necessary but not sufficient: the three Gompertz trajectories are collinear, so joint estimation is credible only under specific design and sample-size conditions.

- The components share an onset shape and overlap in calendar time, so their parameters can be near-collinear even when the identifying contrasts are present.
- Joint recovery requires a phase in which each component varies while the others are held fixed, and an off-drug window long enough for TV to separate from a decaying BR .
- The recoverable region of the design-and-sample-size space is mapped by the pre-registered identifiability diagnostics, not asserted.

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.

Identifiability in the design-contrast sense above is necessary but not sufficient for stable estimation at finite sample sizes. The three Gom-

pertz trajectories share an onset shape and overlap in calendar time, so their parameters can be collinear even when the contrasts that identify them in principle are present. Two conditions make the joint estimation credible: the design must supply at least one phase in which each component varies while the others are held fixed (the blinded-placebo contrast for BR , the open-label-versus-blinded contrast for PB , and an off-drug window for TV), and that off-drug window must be long enough relative to the carryover half-life for TV to separate from a still-decaying BR . Where these conditions are weak the components are jointly identified only up to a near-collinear direction, and the symptom of that is the rank-deficiency the pilot of section 7 encountered when the full phase-augmented formula was fitted at $N = 35$. We therefore do not assert that the three components are always recoverable. The pre-registered identifiability diagnostics of section 7 (convergence rates of the component-matched fit, variance-inflation factors on the component-specific Gompertz parameters, and the sensitivity of the BR estimate to the assumed $\eta(\text{phase})$ multipliers) are the means by which the recoverable region of the design-and-sample-size space is to be mapped, and the framework's participant-level claims should be read as conditional on falling inside that region.

5 Analysis-side methodology

[54]

Four constraints make a component-matched analysis impractical at trial-relevant sample sizes; a phase-augmented LME captures the important part of the gap at minimal cost.

- The question of whether to mirror the DGP in the analysis model is reasonable; the section addresses it directly.
- The four constraints are identifiability, degrees of freedom, convergence fragility, and marginal estimand robustness.
- A phase-by-treatment indicator is the recommended modest extension that captures BR-versus-PB sensitivity without the full parametric cost.

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.

In this section we shall take up the practical question of what analysis model to use. A reasonable reaction to the formal model in section 2 is the following: if we are simulating data with three components plus noise, why is the analysis model written as $Sx \sim \text{bm} + \text{t} + \text{Dbc} + \text{bm}:\text{Dbc}$ with a single random intercept and AR(1) residual correlation? Should we not include three parametric Gompertz components in the analysis, mirroring the DGP, so that the analyst sees the

same structure that generated the data? The question is reasonable. Four constraints make a component-matched analysis impractical for most aggregated N-of-1 trial sizes, and a modest extension (a phase-by-treatment indicator) captures the practically important part of the gap at minimal cost.

5.1 Four reasons not to match the DGP

[55]

The first constraint is identifiability: forcing a three-component model on a design with insufficient contrasts produces unidentified parameters, not component estimates.

- A pure open-label trial cannot distinguish BR from PB ; both rise from zero together on treatment initiation.
- Fitting three components to a one-contrast design produces implausibly tight standard errors that misrepresent inferential precision.
- The analyst must audit whether the design supplies the necessary contrasts before specifying a component-matched model.

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.

The first reason is identifiability. The trial design must supply the contrasts that identify each component, and many designs supply only part of it. The decomposition is identified by phase contrasts (open-label versus blinded for PB , on-drug versus blinded-placebo for BR , off-drug timepoints for TV). A pure open-label trial has none of these contrasts; in such a trial BR and PB both rise from zero on treatment initiation and are statistically indistinguishable from each other, regardless of how the analyst writes the model. Forcing an analysis to fit three components when the design supplies contrasts for only one or two of them does not recover the missing components; it produces unidentified or weakly identified parameter estimates with implausibly tight standard errors that misrepresent the analyst's actual inferential precision.

[56]

The second constraint is degrees of freedom: a component-matched analysis adds approximately a dozen non-linear parameters, inflating moderation estimand variance at $N \in [30, 150]$.

- Gompertz BR (3), PB (3 + expectancy multiplier), and TV (3) each require participant-specific random effects on the maxima.

- At trial-relevant N , the variance of $\hat{\beta}_{bm:D}$ is dominated by participant and timepoint counts.
- Additional parametric structure typically inflates moderation estimand variance more than it reduces bias.

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.

The second reason is degrees of freedom. Each component term costs degrees of freedom that come out of the moderation estimand. The biomarker-moderation question (the core scientific aim of the predictive-biomarker validation programme) is asked through the `bm:D` term. A component-matched analysis adds a Gompertz *BR* (three parameters), a Gompertz *PB* (three parameters plus a phase-specific expectancy multiplier), and a Gompertz *TV* (three parameters), each with participant-specific random effects on the maxima. That is on the order of a dozen new parameters, half of them on a non-linear scale. At trial sample sizes of $N \in [30, 150]$, the variance of $\hat{\beta}_{bm:D}$ is dominated by participant and timepoint counts, and adding parametric structure to absorb residual confounding usually inflates the moderation estimand's variance by more than it reduces bias.

[57]

The third constraint is convergence fragility: component-matched nonlinear mixed models converge below 90% at trial-relevant N , contaminating aggregated analyses.

- Simultaneous Gompertz *BR*, *PB*, and *TV* fitting requires `nlme::nlme`, Bayesian hierarchical, or structural-equation approximations.
- Each is more sensitive to starting values, prone to local optima, and likely to fail on individual trial datasets than the LME analogue.
- Non-converging cells at below-90% convergence rates contaminate power and bias estimates in simulation studies.

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.

The third reason is convergence fragility. Component-matched analyses are non-linear in the parameters and their convergence is problematic in practice. Fitting Gompertz *BR*, *PB*, and *TV* simultaneously requires a non-linear mixed model (`nlme::nlme` with self-starting Gompertz forms), a Bayesian hierarchical fit, or a structural-equation approximation. Each is more sensitive to starting values, more prone to local optima, and more likely to fail to converge on a single trial dataset than the linear-mixed analogue. At trial-relevant

N , the convergence rate of the component-matched fit is typically below 90%, and the non-converging cells contaminate any aggregated analysis.

[58]

The fourth constraint is estimand robustness: the bm:D_{bc} marginal interaction is the right estimand for the biomarker-validation question regardless of the BR-PB split.

- Regulators want to know whether the biomarker predicts treatment response under trial conditions, not a mechanistically decomposed prediction.
- The marginal bm:D_{bc} interaction captures this question whether the moderation is biological or expectation-mediated.
- Component decomposition adds mechanistic precision but is not required for the core biomarker-validation decision.

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.

The fourth reason is that the moderation estimand is sufficiently robust to the BR-versus-PB split for the most common scientific question. For predictive-biomarker validation, the regulator wants to know whether the biomarker predicts treatment response under the trial conditions, not a clean pharmacological-versus-placebo decomposition of that prediction. The simple bm:D_{bc} interaction is the right marginal estimand for the validation question whether the moderation is biological or expectation-mediated.

[59]

The inferential target is stated as an estimand: the population biomarker-by-treatment interaction with explicit intercurrent-event handling, while the pharmacological-component target is more ambitious and inherits the heterogeneity-identifiability caveat.

- The bm:D_{bc} estimand is defined by its population, treatment contrast, endpoint, summary, and intercurrent-event strategy in the ICH E9(R1) sense.
- Dropout is handled by a treatment-policy strategy and residual carryover by the modelled exponential decay.
- Recovering a participant-level pharmacological interaction inherits Senn's caveat that true heterogeneity is separable from within-person variation only under repeated phase contrasts.

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.

It is worth stating the inferential target of the analysis in the vocabulary of the estimands framework⁵³, because the biomarker-validation question is easy to pose ambiguously. The primary estimand is the population-level biomarker-by-treatment interaction, the $bm:D_{bc}$ coefficient: its population is the trial-eligible patient population, its treatment contrast is on-drug versus off-drug exposure summarised through the continuous indicator D_{bc} , its endpoint is the symptom trajectory, its population-level summary is the interaction coefficient, and its intercurrent events are handled by a treatment-policy strategy for dropout and by the modelled exponential decay for residual carryover. The pharmacological-component target β_{bm}^{BR} is a more ambitious estimand: it is the biomarker-by- BR interaction that would obtain with the belief-driven channel held at its off-drug level, identified here through the phase contrasts of section 4 rather than by assumption. We flag explicitly that recovering a participant-level interaction of this kind inherits the difficulty Senn has emphasised for individual treatment effects^{49,54}: a genuine participant-by-component signal is separable from participant-by-period and within-person random variation only when the design supplies repeated phase contrasts, and the framework should be read as estimating these quantities under that condition rather than as dissolving the well-known problem of identifying individual effects from finite within-person data.

5.2 A small extension that usually pays off: phase indicators

[60]

The phase-augmented LME uses `Dbc:phase` and `bm:Dbc:phase` to test whether drug and biomarker-by-drug effects attenuate under blinding.

- `phase` absorbs systematic differences in PB expectancy across trial phases.
- `Dbc:phase` detects whether the drug effect shrinks under blinding, a signature of PB -mediation.
- `bm:Dbc:phase` detects whether the biomarker-by-treatment interaction shrinks under blinding, a signature that it is partly biomarker-by- PB .

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.

The smallest useful step toward component-aware analysis, without paying the costs above, is to add a phase indicator and its interaction with the drug indicator. Schematically,

```
Sx ~ bm + t + Dbc + phase + Dbc:phase
      + bm:Dbc + bm:Dbc:phase
      + (1 | ptID), correlation = corCAR1(...)
```

Here `phase` is a categorical covariate (open-label, blinded, post-crossover) that absorbs the systematic differences in PB expectancy across phases. `Dbc:phase` tests whether the apparent drug effect shrinks under blinding, a signature that the open-label drug effect was partly PB -mediated. `bm:Dbc:phase` tests whether the moderation itself shrinks under blinding, a signature that the biomarker-by-treatment interaction was partly biomarker-by- PB rather than biomarker-by- BR .

[61]

The phase-augmented extension adds at most four parameters yet captures the practically relevant belief-attenuation diagnostic without requiring the Gompertz parametric form.

- Three to four parameters is well within budget at any realistic trial sample size.
- The extension does not separately estimate BR and PB components but identifies the change in apparent drug effect when belief is hedged.
- For most analyses this is the practically relevant question and is answerable without committing to the Gompertz form.

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.

This extension adds three to four parameters at most, well within budget at any realistic N . It does not separately estimate BR and PB as components, but it does identify the change in apparent drug effect that occurs when belief is hedged. For most analyses, this is the practically relevant question, and it can be answered without committing to the Gompertz parametric form. Where both the lumped and the phase-augmented analyses are reported, the additional `Dbc:phase` and `bm:Dbc:phase` terms introduce a small family of belief-attenuation tests, and the attribution inference they support should be treated as a pre-specified secondary analysis with its multiplicity acknowledged rather than as a second bite at the primary interaction test.

5.3 When to fit the full decomposition

[62]

The full decomposition merits the effort only when N is in the hundreds, the design is fully phase-balanced, and the scientific question requires the BR-PB mechanistic split.

- Feasibility conditions: several-hundred N , dense timepoints in every phase, mechanistic BR-PB split required, and informative Bayesian priors available.

- In that regime, `lcmm`, `nlme::nlme` with self-starting Gompertz forms, or `brms` can return participant-specific component estimates.
- Outside that regime the phase-augmented LME is the better tool.

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.

A component-matched analysis merits the effort when sample size is in the several hundreds (a phase-3 trial fielded as aggregated N-of-1, or a meta-analysis pooling multiple smaller trials); the trial design provides timepoints in every phase combination, with enough density per cell to constrain the Gompertz curves; the scientific question requires the BR-PB split (a pharmacology paper claiming a specific mechanism, where ruling out a placebo contribution is the central point); and a Bayesian hierarchical fit with informative priors on the Gompertz parameters is feasible. In that regime, `lcmm`, `nlme::nlme` with self-starting Gompertz forms, or a `brms` hierarchical formulation can return participant-specific *BR*, *PB*, and *TV* component estimates that mirror the DGP. Outside that regime, the linear-mixed analysis with phase indicators is the better tool.

[63]

The DGP-analysis asymmetry is deliberate: simulation uses the rich DGP to characterise how a simpler analysis behaves; the analysis need not mirror the DGP to produce useful inference.

- Simulation controls the DGP and can specify any structure; inference must pay the identifiability cost of every parameter.
- The DGP is a characterisation tool; the analysis is a finite-data inference tool; they have different optimality criteria.
- DGP-matching is not required and is often counterproductive at trial-relevant sample sizes.

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.

The asymmetry between DGP and analysis is not a flaw of the framework; it is a deliberate consequence of the asymmetry between simulation (where the analyst controls the DGP and can specify any structure, however elaborate) and inference (where the analyst must fit a model to a finite dataset and pay the identifiability cost of every additional parameter). Simulation studies use the rich DGP to characterise how a simpler analysis behaves under realistic data generation; they do not require the analysis to match the DGP in order to produce useful inference.

[64]

The exponential-decay carryover indicator is a deliberately parsimonious special case of richer carryover models, and the simulation should benchmark it against a distributed-lag alternative.

- The continuous indicator D_{bc} decays as $\exp(-\lambda t_{sd})$, a one-parameter form tied to a pharmacokinetic half-life.
- Distributed-lag models, the average-period-treatment-effect estimand, and G-estimation relax the single-rate assumption at additional parametric cost.
- The exponential restriction is an assumption, not a finding, and the pre-registered simulation should include a distributed-lag arm to quantify its cost.

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.

A word on the carryover parameterisation is in order, since it is the one place where the analysis model makes a strong functional-form commitment. The continuous drug indicator D_{bc} decays as $\exp(-\lambda t_{sd})$ off drug, a one-parameter exponential governed by the carryover half-life. This is deliberately parsimonious and is best understood as a constrained special case of the more flexible carryover models now available for repeated-measures and N-of-1 data. Liao et al.³¹ model carryover through a Bayesian distributed-lag structure with autocorrelated errors, relaxing the single-rate exponential assumption at the cost of additional parameters; Daza⁵⁵ formalises an average-period treatment-effect estimand built to accommodate autocorrelation, trend, and slow carryover jointly; and G{”a}rtner et al.⁵⁶ benchmark linear models against G-estimation and Bayesian-network alternatives specifically under carryover. We adopt the exponential form because it ties directly to a pharmacokinetic half-life and conserves degrees of freedom for the moderation estimand at trial-relevant sample sizes, but the choice is an assumption rather than a finding. The pre-registered simulation programme of section 7 should accordingly include a distributed-lag analysis arm, so that the cost of the exponential-decay restriction is quantified against the richer alternative rather than asserted.

6 Worked example: information loss without decomposition

6.1 What the lumped analysis estimates

[65]

The lumped treatment-effect estimate has a probability limit equal to the pharmacological component plus belief and

natural-history contributions, so its bias does not vanish as N grows.

- Writing the reduction as $BR + PB + TV - \varepsilon$ and fitting one drug indicator gives $\text{plim } \hat{\beta}_D = m_{BR} + b_{PB} + b_{TV}$.
- b_{PB} and b_{TV} are the belief and natural-history terms that load onto the drug indicator under the chosen contrast; both are generally non-negative for an improving outcome.
- Because the displacement is in the probability limit, a larger one-component trial converges on the same biased value: the failure is identification, not precision.

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.

Before the numerical illustration it is worth establishing algebraically what the one-component analysis actually estimates, because the qualitative failure modes of section 1 are consequences of an identity rather than empirical tendencies that require simulation to demonstrate. Write the within-participant symptom reduction as $R_{it} = BL_i - Y_{it} = BR_{it} + PB_{it} + TV_{it} - \varepsilon_{it}$. The one-component analysis fits a single drug indicator D_{it} ,

$$R_{it} = \alpha + \beta_D D_{it} + g(t) + u_i + e_{it},$$

and reports $\hat{\beta}_D$ as ‘the treatment effect’. Taking the on-drug minus off-drug contrast that β_D targets, and using that the natural-history component is by construction independent of treatment status so that its on/off difference cancels in a within-participant comparison, the probability limit of the estimator is

$$\text{plim } \hat{\beta}_D = m_{BR} + b_{PB} + b_{TV},$$

where $b_{PB} = E[PB | \text{on}] - E[PB | \text{off}]$ is the belief differential between the contrasted phases and b_{TV} collects any natural-history contribution that fails to cancel under the contrast actually used; b_{TV} is zero for a clean on/off within-participant contrast and strictly positive for a baseline-anchored pre-post analysis, in which case $b_{PB} = E[PB]$ and $b_{TV} = E[TV]$. To fix which case is in play below: the one-component analysis used in the worked example and the simulations of this paper is the design-contrast (drug-on/off) form, so $b_{TV} \approx 0$ there and the operative displacement is the belief term b_{PB} ; the baseline-anchored pre-post form, which additionally carries $b_{TV} = E[TV]$, is the worse case retained only for comparison. Both terms are generally non-negative for an improving outcome, so the lumped estimate over-states the pharmacological component m_{BR} . The displacement

is a property of the probability limit and is unchanged as $N \rightarrow \infty$: a larger one-component trial converges on the same biased value. This is the precise sense in which the placebo-attribution failure of section 1 is a problem of identification, not of precision.

[66]

The biomarker slope from a lumped analysis decomposes additively into component-specific slopes, so a biomarker correlated with PB or TV registers as a predictor even with no pharmacological association.

- By linearity of covariance, $\beta_{bm}^{\text{lumped}} = \beta_{bm}^{BR} + w_{PB}\beta_{bm}^{PB} + w_{TV}\beta_{bm}^{TV}$.
- A non-zero β_{bm}^{PB} or β_{bm}^{TV} produces a non-zero lumped slope even when the pharmacological slope β_{bm}^{BR} is exactly zero.
- The result is an algebraic identity under the additive decomposition, not a sample-size-dependent tendency.

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.

The same algebra settles the biomarker-validation failure. The precision-medicine analysis regresses a participant-level response summary on a baseline biomarker B_i . If that summary is the lumped per-participant effect, which estimates $m_{BR,i} + w_{PB}m_{PB,i} + w_{TV}m_{TV,i}$ for design-determined loadings $w_{PB}, w_{TV} \geq 0$, then by the linearity of covariance the fitted biomarker slope has probability limit

$$\beta_{bm}^{\text{lumped}} = \frac{\text{Cov}(m_{BR} + w_{PB}m_{PB} + w_{TV}m_{TV}, B)}{\text{Var}(B)} = \beta_{bm}^{BR} + w_{PB}\beta_{bm}^{PB} + w_{TV}\beta_{bm}^{TV},$$

where $\beta_{bm}^C = \text{Cov}(m_C, B) / \text{Var}(B)$ is the component-specific slope. A biomarker correlated with placebo responsiveness or with the natural-history trajectory, so that β_{bm}^{PB} or β_{bm}^{TV} is non-zero, therefore registers a non-zero lumped slope even when its pharmacological association β_{bm}^{BR} is exactly zero. This is the biomarker-validation failure of section 1, and it too is an identity under the additive decomposition rather than an empirical tendency: no sample size separates the lumped slope from a genuine pharmacological one, because the two coincide in the estimand whenever $w_{PB}\beta_{bm}^{PB} + w_{TV}\beta_{bm}^{TV} \neq 0$.

We make two qualifications explicit, because they bound how much weight the framework can carry. First, both displays are immediate corollaries of the additive decomposition and the linearity of the projection that defines the estimator; they are not a derived result in any deeper sense, and their value is in naming the displacement terms, not in the algebra. Second, they assume the components combine additively. Under the subadditive interaction documented in section 3.3,

where the joint effect of BR and PB falls below their sum, a first-order correction proportional to $\text{Cov}(m_{BR} \cdot m_{PB}, B)$ enters the biomarker slope, and the identities above should be read as the leading behaviour in the additive case rather than as exact under interaction. The empirical magnitude of that correction term is itself an open question that the framework's γ -interaction simulation (section 3.3) is designed to address.

[67]

The bias displaces the estimand rather than inflating variance, so a significance test of the lumped coefficient cannot detect it; the decomposition is what locates it.

- The lumped coefficient is displaced from m_{BR} by $b_{PB} + b_{TV}$ but remains a well-estimated quantity with a valid Wald test of its own nullity.
- Detecting the bias requires comparing $\hat{\beta}_D$ to m_{BR} , a comparison only the decomposition or an isolating phase contrast supplies.
- This is why a rejection-rate summary is the wrong instrument, and accounts for the flat rejection rates across the placebo-strength axis in the section 7.1 pilot.

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.

Two consequences for how the framework should be evaluated follow. First, the quantity that carries the bias is the location of the estimand, $\text{plim } \hat{\beta}_D$, not its sampling variability: the lumped coefficient remains a well-estimated number with a valid Wald test of its own nullity, and that test rejects at the rate dictated by the displaced value. A significance test of the lumped coefficient therefore cannot, in principle, detect the bias, because the bias does not move the coefficient toward zero. Detecting it requires comparing $\hat{\beta}_D$ against m_{BR} , which is exactly the comparison the decomposition, or a phase contrast that isolates m_{BR} , makes available and the lumped analysis does not. Second, and as a corollary for the simulation evidence, a rejection-rate or power summary is the wrong instrument for this bias: the right target is the coefficient itself and its offset from m_{BR} . This is why the pilot of section 7.1 found the biomarker-by-treatment rejection rate essentially flat across the placebo-strength axis even though that axis is precisely what drives the predicted bias; the prediction lives in the mean of $\hat{\beta}_{bm:D_{bc}}$, and a production run that means to test it must read the bias off the coefficient, not off the test.

6.2 A numerical illustration

[68]

The worked example uses two participants with identical lumped responses but opposite BR-PB profiles to show that the one-component analysis cannot distinguish them.

- Participant A has high PB ($m_{PB} = 6$) and moderate BR ($m_{BR} = 4$); Participant B has low PB ($m_{PB} = 1$) and high BR ($m_{BR} = 8$).
- Both participants reach approximately 10–11 points of total improvement by week 8, making them indistinguishable to a one-component analysis.
- The blinded-discontinuation phase creates the diagnostic contrast that exposes the underlying difference.

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.

To illustrate the inferential cost of failing to decompose, we shall work through a simple numerical example. Consider two participants, A and B, enrolled in a sixteen-week N-of-1 trial of prazosin for PTSD. Both are otherwise similar: same age, same baseline severity, same trial design. They differ in two latent traits that the trial does not measure directly. Participant A is a high placebo responder and a moderate pharmacological responder, with true component values $m_{BR}^A = 4$, $m_{PB}^A = 6$, $m_{TV}^A = 1$. Participant B is a low placebo responder and a strong pharmacological responder, with $m_{BR}^B = 8$, $m_{PB}^B = 1$, $m_{TV}^B = 1$.

[69]

Open-label totals are clinically indistinguishable; the one-component analysis estimates a population drug effect of approximately 10 points with no differentiation between A and B.

- Both participants show approximately 10–11 points of improvement and present as clinical successes.
- A t-test or simple LME on open-label change cannot detect the heterogeneity in BR versus PB composition.
- The population drug-effect estimate conflates both pharmacological and placebo contributions.

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.

In the open-label phase (weeks 1 to 8), both BR and PB run at full strength and TV accumulates. By week 8 both participants have reached close to their saturation values:

	<i>BR</i>	<i>PB</i> ($\eta = 1$)	<i>TV</i>	Total improvement
Participant A	4.0	6.0	1.0	11.0
Participant B	8.0	1.0	1.0	10.0

A clinician examining only the totals would conclude that the two participants are responding similarly: both improved by about ten points, both look like clinical successes. A simple t-test on the open-label change would estimate a population drug effect of about ten points and would not distinguish the two participants.

[70]

The blinded-discontinuation phase reveals the BR-PB heterogeneity: Participant A retains 4 points while B collapses to 1.5, exposing the diagnostic contrast.

- Participant A’s high *PB* at $\eta = 0.5$ (3 points) sustains improvement even without active drug.
- Participant B’s low *PB* (0.5 points) and absent *BR* produce near-complete return to baseline.
- The 4.0 versus 1.5 contrast is exactly what the decomposition uses to separate *BR* from *PB*.

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.

In the blinded discontinuation phase (weeks 9 to 12), both participants are silently switched to placebo at the start of week 9. The expectancy multiplier drops from 1.0 to 0.5. For clarity of exposition we take the carryover half-life to be short relative to the four-week blinded window, so that *BR* has effectively cleared by the end of week 12; the more realistic case of a slowly decaying *BR*, modelled by the carryover machinery of section 3.1, attenuates but does not eliminate the contrast below. By the end of week 12, the components are:

	<i>BR</i>	<i>PB</i> ($\eta = 0.5$)	<i>TV</i>	Total improvement
Participant A	0.0	3.0	1.0	4.0
Participant B	0.0	0.5	1.0	1.5

The two participants now look quite different. Participant A still shows a sizeable improvement (four points) under blinded placebo; participant B has nearly returned to baseline (one and a half points). This is the diagnostic contrast that the decomposition exploits. For simplicity the table holds *TV* at its week-8 value; in reality *TV* continues to accumulate over weeks 9 to 12, which shifts both totals by a

1939 common amount and leaves the *BR*-versus-*PB* contrast on which the
1940 example turns unchanged.

1941 [71]

1942 **The one-component model estimates drug at 7.75 points with**
1943 **no principled decomposition, obscuring that A's true *BR* is**
1944 **4 and B's is 8.**

- 1945 • The 7.75 point estimate is a conflated average of true *BR* values
1946 (4.0 and 8.0) plus the placebo wash-out.
- 1947 • The model cannot determine whether A and B differ, nor estimate
1948 the pharmacological drug effect separately.
- 1949 • Placebo and natural-history effects are not identifiable from this
1950 contrast alone.

1951 *Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod*
1952 *tempor incididunt ut labore et dolore magna aliqua.*

1953 A one-component fixed-effects regression of total symptom change on
1954 phase-by-allocation indicators sees:

- 1955 • Open-label means: 11.0 (A), 10.0 (B).
- 1956 • Blinded-placebo means: 4.0 (A), 1.5 (B).
- 1957 • Estimated drug effect: $11.0 - 4.0 = 7.0$ (A), $10.0 - 1.5 = 8.5$ (B).
1958 Population mean: 7.75.
- 1959 • Estimated placebo effect: not identifiable from this contrast.
- 1960 • Estimated natural history: not identifiable from this contrast.

1961 The one-component model has produced a single number conflated
1962 with the placebo response that washes out under blinding. It cannot
1963 say whether the drug effect is the same for participants A and B; it
1964 estimates 'drug' at 7.75 points on average, which is an average of the
1965 true 4.0 (A) and 8.0 (B) mixed with the placebo wash-out, with no
1966 principled way to decompose further.

1967 [72]

1968 **The full decomposition recovers all six component estimates**
1969 **by exploiting phase differential expectancy and the blinded-**
1970 **placebo contrast.**

- 1971 • The LME returns $\hat{m}_{BR}^A = 4.0$, $\hat{m}_{BR}^B = 8.0$, $\hat{m}_{PB}^A = 6.0$, $\hat{m}_{PB}^B =$
1972 1.0 at the population point estimate.
- 1973 • Participant-level estimates have meaningful uncertainty at trial-
1974 relevant N ; the simulation study characterises this.
- 1975 • The phase differential ($\eta = 1$ vs. $\eta = 0.5$) is the identification
1976 mechanism for the *PB* parameter.

1977 *Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod*
1978 *tempor incididunt ut labore et dolore magna aliqua.*

A linear-mixed-effects analysis with the full BR-PB-TV decomposition, fitted to the same data, can recover the individual components by exploiting the differential expectancy between phases and the on-drug-versus-blinded-placebo contrast. After fitting, the model returns $\hat{m}_{BR}^A = 4.0$, $\hat{m}_{BR}^B = 8.0$, $\hat{m}_{PB}^A = 6.0$, $\hat{m}_{PB}^B = 1.0$, $\hat{m}_{TV}^A = \hat{m}_{TV}^B = 1.0$ (at the population point estimate; participant-level estimates have meaningful uncertainty at N in the trial-relevant range, which the simulation study below characterises).

[73]

The one-component model produces one biased average; the three-component model produces six clinically actionable quantities, making biomarker validation possible.

- The lumped estimate of 7.75 points is a biased average of $BR_A = 4$ and $BR_B = 8$ confounded with placebo wash-out.
- The decomposition supplies individual BR , PB , and TV estimates enabling per-participant clinical decisions.
- Predictive-biomarker validation requires the decomposition because biomarker-by- BR is the target estimand.

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.

The information loss from lumping the components is not abstract. We may compare the inferences directly:

Question	One-component answer	Three-component answer
Does the drug work for A?	~7.75 points (lumped).	$BR_A = 4.0$, distinct from her larger placebo response.
Does the drug work for B?	~7.75 points (lumped).	$BR_B = 8.0$, twice the magnitude of A.
Are A and B different responders?	No discernible difference.	B has 2x A's BR but 1/6 of A's PB .
Is a biomarker predicting BR useful here?	Cannot answer.	Yes if the biomarker correlates with $\hat{m}_{BR}$.
Should A be continued long-term?	Cannot say.	Probably yes ($BR = 4$ is real and pharmacological), but the larger placebo component will not persist if expectations decay.

Question	One-component answer	Three-component answer
Population mean drug effect?	7.75 points (biased).	$6.0 = (4.0 + 8.0)/2$ (correct mean of BR).

2001 The one-component model has produced one biased average where the
 2002 three-component model has produced six clinically actionable quanti-
 2003 ties. Predictive-biomarker validation, which is a central scientific aim
 2004 of the trial design family considered here, becomes impossible without
 2005 the decomposition.

2006 [74]

2007 **A population-level example shows that cohort differences in**
 2008 **open-label totals driven entirely by PB produce apparent**
 2009 **non-replicability that the decomposition resolves.**

- 2010 • Cohort 1 (naive, $m_{PB} = 7$) reports 13-point improvements; Co-
 2011 hort 2 (sceptical, $m_{PB} = 2$) reports 8 points; true $m_{BR} = 5$ in
 2012 both.
- 2013 • A one-component analysis would falsely conclude the drug works
 2014 less well in Cohort 2 and trigger searches for confounders.
- 2015 • The decomposition returns $\hat{m}_{BR} = 5$ in both cohorts, attributing
 2016 the difference correctly to PB heterogeneity.

2017 *Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod*
 2018 *tempor incididunt ut labore et dolore magna aliqua.*

2019 A second example at the population level illustrates the same problem.
 2020 Consider two cohorts of N-of-1 participants enrolled in methodologi-
 2021 cally identical trials of the same drug. Cohort 1 (early trial, naive pa-
 2022 tients) has mean $m_{BR} = 5$, mean $m_{PB} = 7$, mean $m_{TV} = 1$. Cohort 2
 2023 (later trial, patients who have had multiple prior failed treatments and
 2024 are sceptical) has mean $m_{BR} = 5$, mean $m_{PB} = 2$, mean $m_{TV} = 1$.
 2025 By construction the drug works equally well in both cohorts. But the
 2026 open-label totals differ substantially: cohort 1 reports thirteen-point
 2027 improvements on average, cohort 2 reports eight-point improvements.
 2028 A one-component analysis comparing the two cohorts would conclude
 2029 that the drug works less well in cohort 2 and might trigger a search
 2030 for confounders or sub-population effects. None of this is real; the dif-
 2031 ference is entirely in PB , specifically in the population baseline level
 2032 of placebo responsiveness. The three-component decomposition, ap-
 2033 plied to either cohort, returns $\hat{m}_{BR} = 5$ in both. Unfortunately, this
 2034 is the kind of apparent non-replicability that pharmacology has spent
 2035 the last decade struggling to navigate; the decomposition is one of the
 2036 formal tools that makes it tractable.

7 Simulation study

[75]

The simulation study follows ADEMP and quantifies bias and identifiability properties of three analysis strategies across a prazosin-PTSD-calibrated parameter grid.

- The Morris (2019) ADEMP framework structures the study: Aims, DGP, Estimands, Methods, Performance measures.
- The pre-registration is at `analysis/scripts/component-decomposition/00-ademp-pre-registration`.
- Three analysis strategies (one-component, phase-augmented, full decomposition) are compared across the parameter grid.

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.

The simulation study reported and pre-registered in this section is structured under the⁵⁷ ADEMP framework, specifying Aims, Data-generating mechanisms, Estimands, Methods, and Performance measures. The pre-registered plan is at `analysis/scripts/component-decomposition/00-ademp-pre-registration.md`. A Monte Carlo simulation calibrated to the prazosin/PTSD reference parameters quantifies the bias and identifiability properties of the three analysis strategies described in section 5 across a parameter grid.

7.1 Pilot validation of analysis machinery (100 replicates, 6 cells)

[76]

The pilot is a structural check of the analysis pipeline, not a substantive empirical result.

- The realised Study A run (section 7.3) is the empirical core of the manuscript; this pilot precedes it.
- The pilot confirms only that data generation, model fitting, and summary output run without error.
- No bias or power claims are derived from the 100-replicate pilot.

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.

The production simulation programme is the empirical core of this manuscript; its first realised run, Study A, is reported in section 7.3, while the fuller pre-registered grid (section 7.2) remains partly outstanding. The pilot reported here precedes both and confirms only that the analysis machinery runs end-to-end.

[77]

The pilot executed a 6-cell, 100-replicate run with perfect convergence, confirming the pipeline is operational.

- The grid crossed three $pb_strength$ levels against two analysis strategies.
- Convergence was 1.0 across all six cells under 8-core `mclapply`.
- Output is checkpointed at `analysis/data/quick-sim/`; no inferential claims are drawn.

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.

A 100-replicate-per-cell pilot was run at the reference Hybrid OL+BDC design, $N = 35$, on the 6-cell slice $pb_strength \in \{0, 6.5, 13\} \times analysis \in \{\text{one-component, phase-augmented}\}$, using 8-core `parallel::mclapply`. The convergence rate was 1.0 across all six cells. Per-replicate output is checkpointed at `analysis/data/quick-sim/06-decomp-replicates.rds` and the Morris-format cell-level summary at `analysis/data/quick-sim/06-decomp-summary.txt`. The exploratory summaries are diagnostic only. The one-component `bm:Dbc` test showed power 0.95, 0.93, and 0.83 across $pb_strength \in \{0, 6.5, 13\}$, while the phase-augmented analysis showed 0.65, 0.65, and 0.61; the mean estimated $\hat{\beta}_{bm:D_{bc}}$ was essentially flat within each analysis (about -2.25 for the one-component and -2.21 for the phase-augmented fit). At 100 replicates the Monte Carlo standard error on a power estimate near 0.7 is about 0.046, so these figures cannot by themselves separate the two analyses; their interpretation, including why the phase-augmented analysis loses its bias-detection contrast at this N , is resolved by the production run of section 7.3. No confirmatory bias or power claim is drawn from the pilot, and all substantive claims are deferred to the production programme below.

[78]

Two pilot findings revise the production specification: rank-deficiency forces a formula change, and bias must be read from coefficient means, not rejection rates.

- At $N = 35$, rank-deficiency dropped the $D_{bc}:\text{phase}$ term, eliminating the key biomarker-by-treatment-by-phase contrast.
- The predicted bias manifests in the mean of $\hat{\beta}_{bm:D_{bc}}$, not in the Wald-test rejection rate.
- The production programme addresses both by using larger N , a fuller design, and coefficient-mean reporting.

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.

Two design-level observations from the pilot inform the production

specification. First, at $N = 35$ the rank-deficiency of the full phase-augmented formula required dropping the D_{bc} :phase interaction term, removing the precise contrast that would have isolated the biomarker-by-treatment-by-phase signal the framework is designed to detect. Second, the predicted bias of the one-component analysis arises in the regression coefficient $\hat{\beta}_{bm:D_{bc}}$, not in the indicator-level rejection rate, so power on the biomarker-by-treatment Wald test is not the right summary statistic to detect the framework's predicted bias. The production programme below addresses both observations through larger N , a richer trial design that admits the full phase-augmented formula, and reporting of the coefficient mean against the implied true value rather than the rejection rate alone.

7.2 Production-simulation specification

[79]

The production programme follows the ADEMP framework with four deliverables; the realised Study A run in section 7.3 is the first of these.

- The deliverables are interaction-estimate bias, full-decomposition identifiability, type I error and power, and participant-level signal recovery.
- Sample sizes $N \in \{35, 70, 100, 150\}$ cross a *pb_strength* grid; type I cells target MCSE 0.003 at 5,000 replicates.
- Driver scripts, checkpointed outputs, and the ADEMP pre-registration reside under `analysis/scripts/component-decomposition/`.

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.

The production programme follows the⁵⁷ ADEMP framework and targets four deliverables: the bias of $\hat{\beta}_{bm:D_{bc}}$ for the one-component, phase-augmented, and full-decomposition analyses across $N \in \{35, 70, 100, 150\}$ and a *pb_strength* grid; identifiability metrics for the full-decomposition fit (convergence, variance-inflation factors, and sensitivity of the *BR* estimate to the η (phase) multipliers); type I error and power for the interaction test under an η -mismatch, *TV*-magnitude, and dropout sensitivity grid, with null cells at $n_{\text{reps}} = 5,000$ for MCSE 0.003; and the participant-level interaction signal recovered as a function of (m_{BR}, m_{PB}, m_{TV}) , which the lumped-total regression does not preserve. The `tidyverse` implementation collection drives the grid; driver scripts, checkpointed cell-level summaries, per-replicate output, and the committed ADEMP pre-registration all reside under `analysis/scripts/component-decomposition/`. The realised Study A run reported next used 1,000 replicates per alternative cell.

7.3 Study A results: bias and power at 1,000 replicates per alternative cell

[80]

At 1,000 replicates per alternative cell the one-component estimate of $\hat{\beta}_{bm:D_{bc}}$ is essentially unbiased across the entire (m_{PB}, m_{TV}) grid, contradicting the framework's prediction of a placebo-strength-driven bias.

- The true biomarker-by- BR effect is -2.25 ; mean absolute bias is 0.016 at $N = 35$, 0.016 at $N = 70$, 0.006 at $N = 100$, and 0.006 at $N = 150$, in every case within one Monte Carlo SE of zero.
- Bias shows no monotone trend across $m_{PB} \in \{0, 1, 3, 6, 10\}$ or $m_{TV} \in \{-1, 0, 1, 2\}$; it behaves as sampling noise around zero.
- This is the predicted consequence of the identity in section 6.1: the DGP couples the biomarker to BR only, so $\beta_{bm}^{PB} = \beta_{bm}^{TV} = 0$ and the lumped slope is unbiased by construction.

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.

The production Study A run (1,000 replicates per alternative cell, 5,000 per null cell, 80 alternative cells crossing $m_{PB} \in \{0, 1, 3, 6, 10\}$, $m_{TV} \in \{-1, 0, 1, 2\}$, and $N \in \{35, 70, 100, 150\}$, plus four null cells) returns a result that does not support the framework's central bias prediction. The data-generating process couples the baseline biomarker to the BR component only, through the on-drug biomarker- BR covariance channel, so the true biomarker-by-treatment effect is the biomarker-by- BR slope, $\beta_{bm:D_{bc}} = -c_{bm} \sigma_{BR} / \sigma_{bm} = -2.25$, and the biomarker is by construction uncorrelated with PB and TV . Under the one-component analysis the mean estimated $\hat{\beta}_{bm:D_{bc}}$ tracks this value across the whole grid: the mean absolute bias is 0.016 at $N = 35$, 0.016 at $N = 70$, 0.006 at $N = 100$, and 0.006 at $N = 150$, in every case within one Monte Carlo standard error (respectively about 0.028 , 0.019 , 0.009 , and 0.007) of zero, with no monotone trend across m_{PB} or m_{TV} . The predicted placebo-attribution bias of the biomarker-by-treatment estimate does not appear.

This is not a failure of the simulation; it is the behaviour the identity of section 6.1 requires. That identity writes the lumped biomarker slope as $\beta_{bm}^{BR} + w_{PB} \beta_{bm}^{PB} + w_{TV} \beta_{bm}^{TV}$. With the biomarker coupled to BR alone, $\beta_{bm}^{PB} = \beta_{bm}^{TV} = 0$, so the lumped slope equals the pharmacological slope no matter how large m_{PB} or m_{TV} grows. Study A varies the magnitudes of the unmodelled components, which the identity shows to be the innocuous axis; the axis that produces biomarker-validation bias is the biomarker's correlation with PB or TV , which this DGP holds at zero. The clean null is therefore a confirmation of the identity

and a correction to the looser claim of section 1 that large PB or TV by itself biases biomarker validation. It does not; biomarker-component coupling does.

[81]

The phase-augmented analysis provides no bias reduction and incurs a substantial power penalty, so it is dominated under this uncontaminated DGP.

- Mean power at $N = 35$ is 0.35 (phase-augmented) versus 0.59 (one-component); at $N = 70$, 0.62 versus 0.90; both reach approximately 1.0 by $N = 100$.
- Phase-augmented mean absolute bias is no smaller than one-component at any N (0.028 versus 0.016 at $N = 35$).
- Convergence is approximately 100% for both analyses at every N (minimum 0.998), so the rank-deficiency that degraded the $N = 35$ pilot does not recur here.

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.

The comparison between analyses is equally clear. The phase-augmented analysis is not less biased than the one-component analysis at any sample size (its mean absolute bias is 0.028 versus 0.016 at $N = 35$ and is comparable elsewhere), and it is markedly less powerful for the biomarker-by-treatment test: mean power across the grid is 0.35 versus 0.59 at $N = 35$ and 0.62 versus 0.90 at $N = 70$, with both analyses reaching essentially full power by $N = 100$. Because the DGP contains no biomarker- PB contamination for the phase contrast to remove, the extra phase-by-treatment terms add estimation cost without buying any bias reduction, and the analysis is dominated. Convergence was approximately complete for both analyses at every sample size (minimum convergence rate 0.998), so the rank-deficiency that forced dropping the `Dbc:phase` term in the $N = 35$ pilot, which we had advanced as the explanation for the pilot's inverted power ranking, is not operative in the production run: the inverted ranking persists with the full formula fitted and converged. The pilot's power inversion was therefore a genuine feature of this DGP, not an artifact of rank-deficiency.

Calibration is adequate for both analyses. Nominal-95% confidence-interval coverage of $\beta_{bm:D_{bc}}$ ranges from 0.96 to 0.99 for the one-component analysis (slightly conservative) and from 0.94 to 0.98 for the phase-augmented analysis. Type I error at the four null cells is controlled, ranging from 0.031 to 0.056 for the one-component analysis and 0.038 to 0.049 for the phase-augmented analysis; the one-component estimate at $N = 100$ (0.056, MCSE 0.003) sits 1.9 standard errors above nominal but is within the simulation's Monte Carlo

uncertainty.

[82]

Study A confirms the identity but cannot exhibit the biomarker-validation failure, which requires a DGP that couples the biomarker to PB ; that corrected cell is the necessary next run.

- As pre-registered, Study A varies component magnitude with the biomarker fixed to BR , so it probes an axis the identity predicts to be null.
- Demonstrating the predicted contamination requires varying β_{bm}^{PB} , a biomarker- PB covariance, at fixed m_{PB} .
- On current evidence the phase-augmented analysis is not recommended for this design; it should be re-evaluated only under a contaminated-biomarker DGP.

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.

Two conclusions follow for the programme. First, the result sharpens rather than supports the section 1 framing: the biomarker-validation failure is real as an identity, but it is driven by the biomarker's coupling to PB or TV , not by the magnitude of those components, and Study A as pre-registered varies the latter while holding the former at zero. The decisive cell, which we did not run here, varies the biomarker- PB covariance β_{bm}^{PB} at fixed m_{PB} ; the identity predicts the lumped bias to scale with $w_{PB}\beta_{bm}^{PB}$, and that is the cell that would confirm or refute the practical importance of the failure mode. Second, on the present evidence the phase-augmented analysis carries a real power cost without a compensating bias reduction under an uncontaminated biomarker, so it should not be recommended as a default for this design; its value, if any, is confined to the contaminated-biomarker regime that the corrected Study A would probe. We report this negative result in full because it bounds the claims the framework can currently support.

8 Application to predictive-biomarker validation

[83]

The precision-medicine question – whether a biomarker predicts pharmacological response – is answered by the lumped total only when the biomarker is uncorrelated with PB and TV ; otherwise it requires decomposing the response.

- The question is whether the biomarker predicts the BR component, not the lumped total.

- The PATH statement formalises this as the target estimand for predictive biomarker validation.
- The BR-PB-TV decomposition is the mechanism that separates the pharmacological signal when biomarker-component coupling cannot be ruled out.

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.

The motivating clinical question for the trial designs in this project is whether a baseline biomarker predicts treatment response, and specifically whether the biomarker predicts the pharmacological component of treatment response. This is the precision-medicine question, formalised in the PATH statement¹⁴. Whether it depends on the BR-PB-TV decomposition is itself conditional: by the identity of section 6.1, the lumped biomarker slope equals the pharmacological slope when the biomarker is uncorrelated with *PB* and *TV*, and departs from it only by the biomarker's covariance with those components. The decomposition is therefore the mechanism the question requires precisely in the regime where that covariance is non-negligible.

[84]

A lumped-total biomarker regression conflates pharmacological and placebo-belief prediction, producing a stratification biomarker that mis-identifies responders.

- $\hat{\beta}_{bm}$ from the lumped regression contains both pharmacological and placebo-belief prediction in unknown proportions.
- If *PB* covaries with the biomarker distribution, flagged 'high responders' include high-*PB* responders who will not benefit pharmacologically.
- Prospective application of such a biomarker would systematically mis-stratify patients for prescription decisions.

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.

A direct regression of total response on baseline biomarker

```
total_response = beta_0 + beta_bm * biomarker + error
```

estimates the biomarker's correlation with whatever the lumped total contains. If *PB* varies systematically across the biomarker distribution, because the biomarker is correlated with optimism, suggestibility, or treatment-seeking history, then $\hat{\beta}_{bm}$ contains a placebo-prediction component inseparable from the pharmacological-prediction component. The resulting biomarker, applied prospectively, would mis-stratify patients: those flagged as 'high responders' would include both true high-*BR* responders and high-*PB* responders, and

the latter would not benefit from prescription decisions made on a pharmacological basis.

[85]

A valid pharmacological predictor requires a non-zero $\hat{\beta}_{bm}^{BR}$; significant $\hat{\beta}_{bm}^{PB}$ or $\hat{\beta}_{bm}^{TV}$ indicate a psychological or prognostic biomarker, not a pharmacological one.

- The sole criterion for pharmacological prediction is a meaningfully non-zero BR -component regression coefficient.
- A biomarker correlating only with PB selects placebo responders, not drug responders.
- A biomarker correlating only with TV is prognostic but not predictive in the pharmacological sense.

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.

The right approach is to regress each component on the biomarker:

$$\begin{aligned} BR &= \beta_0^{BR} + \beta_{bm}^{BR} \cdot \text{biomarker} + \varepsilon^{BR}, \\ PB &= \beta_0^{PB} + \beta_{bm}^{PB} \cdot \text{biomarker} + \varepsilon^{PB}, \\ TV &= \beta_0^{TV} + \beta_{bm}^{TV} \cdot \text{biomarker} + \varepsilon^{TV}. \end{aligned}$$

A biomarker is a useful pharmacological predictor if and only if $\hat{\beta}_{bm}^{BR}$ is meaningfully non-zero. Significant $\hat{\beta}_{bm}^{PB}$ or $\hat{\beta}_{bm}^{TV}$ are diagnostic findings about the biomarker, indicating that it is partly psychological or correlates with natural-history characteristics, but they are not the validation of pharmacological prediction. A biomarker that correlates only with PB would be valuable for selecting placebo responders in run-in designs but worthless for predicting pharmacological efficacy; a biomarker that correlates only with TV would be a prognostic marker, not a predictive one.

[86]

When biomarker-component coupling cannot be ruled out, the decomposition is the mechanism by which the precision-medicine question is posed; when it can, the lumped analysis already answers it.

- Where the biomarker may correlate with PB or TV , separating BR from them is what prevents validation from conflating pharmacological and non-pharmacological prediction.
- The phase-augmented `bm:Dbc:phase` interaction is a small- N proxy, but Study A shows it buys nothing under an uncontami-

2359 nated biomarker, so it is warranted only when contamination is
2360 plausible.

- 2361 • The full decomposition recovers component-specific regressions
2362 explicitly when sample size permits.

2363 *Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod*
2364 *tempor incididunt ut labore et dolore magna aliqua.*

2365 The decomposition is therefore the mechanism by which the precision-
2366 medicine question is posed whenever the candidate biomarker may cor-
2367 relate with PB or TV . Where the biomarker can be assumed orthog-
2368 onal to the non-pharmacological components, the identity of section
2369 6.1 shows the lumped analysis already targets the biomarker-by- BR
2370 estimand, and the simulation of section 7.3 confirms it recovers that
2371 estimand without bias. The phase-augmented linear-mixed analysis
2372 introduced in section 5 retains a useful subset of the decomposition
2373 machinery at small N (the `bm:Dbc:phase` interaction tests whether the
2374 biomarker-by-treatment effect attenuates under blinding, the practical
2375 proxy for biomarker-by- PB contamination), but Study A shows this
2376 proxy earns its degrees of freedom only when such contamination is
2377 present: under an uncontaminated biomarker it reduced no bias and
2378 lost power. The full decomposition recovers the component-specific re-
2379 gressions explicitly when N permits and when the scientific question
2380 warrants the cost.

2381 9 Discussion

2382 [87]

2383 **The three-component framework reframes treatment re-**
2384 **sponse as a structured sum of causally distinct contributions,**
2385 **and the present work maps when acting on that decomposi-**
2386 **tion actually changes inference.**

- 2387 • The framework specifies which trial-design contrasts identify
2388 which components.
- 2389 • Acting on the decomposition is valuable conditionally: for attri-
2390 bution estimands, and for biomarkers that may correlate with PB
2391 or TV , but not for a biomarker-by-treatment interaction with a
2392 biomarker orthogonal to them.
- 2393 • The target estimand for biomarker validation is the biomarker-
2394 by- BR association; whether a lumped analysis already recovers
2395 it depends on the biomarker’s coupling to PB and TV .

2396 *Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod*
2397 *tempor incididunt ut labore et dolore magna aliqua.*

2398 The three-component decomposition is the formal expression of the
2399 claim that ‘treatment response’ in an aggregated N-of-1 trial is not a

single quantity but a sum of three causally distinct contributions, each with its own pharmacological, psychological, or epidemiological interpretation. What the present work establishes, analytically in section 6.1 and empirically in section 7.3, is that the inferential value of acting on this decomposition is conditional rather than universal. The identity shows that the one-component estimator is displaced from its pharmacological target only by the components' loading onto the contrast used, and, for a biomarker slope, only by the biomarker's covariance with PB and TV . Where those terms are zero, the design-contrast one-component analysis is already unbiased, and the production simulation confirms it: with a biomarker coupled to BR alone, the one-component biomarker-by-treatment estimate is unbiased across the whole $PB-TV$ grid and the phase-augmented analysis is dominated. The framework's contribution is therefore best read as a map of when decomposition changes inference, not as a blanket prescription to decompose. It makes the trial-design conditions explicit, identifies the attribution and contaminated-biomarker estimands for which decomposition is needed, and clarifies the target estimand for biomarker validation (the biomarker-by- BR association, not the lumped total).

[88]

Four open questions bound the framework's current scope: the simulation results are pending, the η multipliers require sensitivity analysis, non-monotone natural history is not yet handled, and participant-level recovery inherits the identifiability caveats of individual treatment effects.

- Quantitative bias-versus-power trade-offs are conjectures until the section 7 simulation is run.
- The η (phase) expectancy-modulation parameters are modelling assumptions requiring explicit sensitivity analysis.
- Cyclical or biphasic TV trajectories exceed the Gompertz form and require flexible or covariate-augmented specifications.
- Participant-level component recovery rests on separating participant-by-component signal from participant-by-period and participant-by-phase variation, the identifiability caveat Senn raises for individual treatment effects.

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.

Four open questions remain and warrant acknowledgement. First, the simulation study described in section 7 is forthcoming; the present manuscript reports the framework rather than its empirical evaluation, and the quantitative bias-versus-power trade-offs across analysis strategies are conjectures until that work is run. Second, the η (phase) multipliers used to parameterise the expectancy modulation of PB

are themselves modelling assumptions, and a sensitivity analysis varying η across plausible values is mandatory before any clinical conclusion is drawn from the framework's outputs. Third, the framework as specified treats the three components as monotone Gompertz trajectories; cyclical or biphasic TV patterns require either flexible TV specifications or explicit covariates beyond the Gompertz form, and the boundary between 'flexible-Gompertz' and 'covariate-augmented' specifications is not yet sharply drawn. Fourth, the participant-level component estimates the framework targets inherit the identifiability caveat that⁵⁴ and⁴⁹ raise for individual treatment effects more generally: within-participant component recovery rests on separating a true participant-by-component signal from participant-by-period and participant-by-phase variation, and this separation is credible only when the design supplies repeated phase contrasts and the cross-component covariance structure is not pathological. The framework should be read as estimating these quantities under those assumptions, not as dissolving the well-known difficulty of estimating individual effects from finite within-person data.

[89]

Two companion documents extend the manuscript: a pedagogical narrative and a Gompertz-family sensitivity evaluation.

- The pedagogy document at `docs/24-component-decomposition-pedagogy.md` provides worked examples for first-time readers.
- The Gompertz evaluation at `analysis/report/07-gompertz-evaluation/` quantifies trajectory-specification sensitivity.
- Together they address the intuitive and parametric dimensions of the framework that the present manuscript treats briefly.

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.

The companion pedagogical document at `docs/24-component-decomposition-pedagogy.md` develops the framework with worked examples and an extended intuition narrative for readers approaching the material for the first time. The companion methodological evaluation of the Gompertz family at `analysis/report/07-gompertz-evaluation/` quantifies the sensitivity of the framework's inference to the specific parametric form chosen for the component trajectories.

[90]

The components exist in any longitudinal treatment response, but participant-level identifiability requires a within-subject belief-by-exposure contrast that not every placebo-controlled design supplies.

- The three-component structure is a property of the data-generating process of any repeated-measures treatment study; its identifiability is design-specific.
- Parallel-group placebo control identifies the population-mean PB but not a participant-level PB , and supplies no within-subject belief contrast; only designs with such a contrast (blinded or open-hidden discontinuation, randomised withdrawal, crossover, hybrid N-of-1) identify the participant-level decomposition the biomarker question needs.
- The broad claim is therefore scoped to designs supplying a within-subject belief-by-exposure contrast, not to placebo-controlled longitudinal trials in general.

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.

A final point concerns the scope of the decomposition, where it is essential to distinguish the existence of the components from their identifiability. The claim that an observed longitudinal treatment response is the sum of a pharmacological, a belief-driven, and a natural-history component is a statement about the data-generating process of any repeated-measures treatment study, and in that sense the decomposition is not specific to N-of-1 trials. Identifiability, by contrast, is design-specific, and the distinction between population-level and participant-level recovery is decisive for the biomarker application. A standard parallel-group placebo-controlled trial supplies a between-arm placebo contrast that identifies the population-mean belief response but not a participant-level PB , and it contains no within-subject contrast in which belief varies independently of pharmacology; the participant-level biomarker-by- BR slope that the precision-medicine question targets is therefore not identified from such a design, no matter how large it is. The designs that do supply the required within-subject belief contrast are the blinded or open-hidden discontinuation and randomised-withdrawal designs^{51,52}, the crossover with on/off periods, and the hybrid N-of-1 design developed here; the pharmacometric disease-progression-plus-placebo models from which the framework descends¹⁻³ operate at the group level and recover population-mean components only. We therefore advocate the decomposition as a default lens for longitudinal designs that supply a within-subject belief-by-exposure contrast, with the aggregated N-of-1 case the most demanding instance because it must recover the components from a single participant's trajectory. We do not claim participant-level identifiability from parallel-group placebo control alone, and where only a between-arm placebo contrast is available the framework's claims are restricted to the population mean.

10 Conclusions

[91]

Treatment response in aggregated N-of-1 trials is the sum of three causally distinct components, and the bias from conflating them is governed by the estimand and the biomarker's coupling to the components, not by their raw magnitude.

- The three components are BR (pharmacological), PB (placebo-belief), and TV (natural history).
- Each has its own biological mechanism, identifiability conditions, and clinical interpretation.
- The lumped pre-post treatment effect is biased by the PB and TV contributions; the biomarker-by-treatment slope is biased only by the biomarker's covariance with PB and TV , as Study A confirms.

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.

In conclusion, three points are to be emphasised. First, treatment response in aggregated N-of-1 clinical trials is the sum of three causally distinct components: pharmacological action (BR), placebo-belief response (PB), and time-variant natural history (TV). Each component has its own biological mechanism, its own identifiability conditions, and its own clinical interpretation. Whether conflating them biases a given analysis is, however, governed by the omitted-variable-bias identity of section 6.1, not by the raw magnitude of PB and TV : the lumped pre-post treatment effect is displaced by the PB and TV contributions, whereas the biomarker-by-treatment slope is displaced only by the biomarker's covariance with those components. Study A bears this out: varying PB and TV magnitude with a biomarker coupled to BR alone left the one-component estimate unbiased across the entire grid.

[92]

What matters is the design contrast, not the parametric model: the trial must supply the drug-on/off and belief contrasts, but adding parametric component terms is not free and is not always worth it.

- Unique component identification requires phase contrasts built into the trial design; the one-component analysis already exploits the drug-on/off contrast.
- Under an uncontaminated biomarker, the phase-augmented analysis adds no bias reduction and loses power, so it is not recommended as a default.

- 2570 • The phase-augmented and full component-matched analyses are
2571 warranted for attribution estimands and for biomarkers that may
2572 correlate with PB or TV , where the extra structure does work.

2573 *Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod*
2574 *tempor incididunt ut labore et dolore magna aliqua.*

2575 Second, the operative requirement is the design contrast, not the
2576 parametric analysis model. The trial design must include the drug-
2577 on/off and belief contrasts that identify the components, and the one-
2578 component analysis already exploits the drug-on/off contrast through
2579 its D_{bc} term. Adding parametric component structure on top is not
2580 free: under an uncontaminated biomarker the phase-augmented anal-
2581 ysis reduced no bias and lost substantial power, so it should not be
2582 adopted as a default. Its value, and that of the full component-
2583 matched fit, is confined to the estimands where the extra structure
2584 does work, namely the attribution of response to pharmacology versus
2585 belief and the analysis of biomarkers that may themselves correlate
2586 with PB or TV .

2587 [93]

2588 **For predictive-biomarker validation the decomposition mat-**
2589 **ters precisely when the candidate biomarker may correlate**
2590 **with PB or TV ; this is the case the decisive simulation still**
2591 **has to probe.**

- 2592 • A biomarker that predicts the lumped total is a pharmacological
2593 predictor only if it is uncorrelated with PB and TV ; otherwise
2594 the lumped slope is contaminated.
- 2595 • Regulatory and prescribing decisions therefore require separating
2596 BR from PB and TV whenever biomarker-component coupling
2597 cannot be ruled out.
- 2598 • Study A established the uncontaminated baseline; the
2599 contaminated-biomarker cell is the planned empirical test
2600 of the failure mode.

2601 *Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod*
2602 *tempor incididunt ut labore et dolore magna aliqua.*

2603 Third, for predictive-biomarker validation the decomposition earns
2604 its place precisely when the candidate biomarker may correlate
2605 with PB or TV . A biomarker that predicts the lumped total is
2606 a pharmacological predictor only when it is uncorrelated with the
2607 non-pharmacological components; where that cannot be assumed,
2608 the lumped slope mixes pharmacological and non-pharmacological
2609 prediction, and the regulatory and prescribing decisions that motivate
2610 the trial require the separation the decomposition supplies. The
2611 empirical demonstration of this failure mode, a data-generating

process that couples the biomarker to PB , is the decisive cell that Study A did not include and that remains the natural next step; the present run establishes the uncontaminated baseline against which it should be compared.

11 Conflict of interest

The authors declare no conflicts of interest.

12 Funding

This work received no specific funding.

13 Data availability statement

The data and code that support the findings of this study are openly available in the pmsimstats-ng project repository. Per-replicate Monte Carlo outputs and ADEMP-compliant cell-level summaries are versioned under `analysis/data/`; simulation drivers are at `analysis/scripts/`. Exact paths for this manuscript are listed in the Reproducibility section below.

14 Reproducibility

[94]

The manuscript is rendered inside a Docker container with locked package versions; the full software environment is pinned by Dockerfile and renv.lock.

- R 4.4.x in the zzcollab Docker container provides the execution environment.
- Simulation packages are `nlme`, `parallel`, `data.table`, `furrr`, and `purrr`.
- The manuscript build uses `rmarkdown` and `bookdown`.

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.

This manuscript was rendered with R 4.4.x inside the project's zzcollab Docker container (image hash recorded in `Dockerfile`; package set captured in `renv.lock`). Principal packages used by the simulation pipeline are `nlme` (continuous-time linear-mixed analysis with `corCAR1` residual correlation), `parallel` (8-core `mclapply` for parameter-grid sweeps), `data.table` (the original-extended simulation collection), `furrr` and `purrr` (the tidyverse simulation collection), and `rmarkdown` plus `bookdown` for the manuscript build.

[95]

Reproducibility is ensured by a master seed of 20260507 with per-replicate seeds derived by index addition.

- `set.seed(20260507)` is called at the top of each replicate loop.
- `parallel::mc.set.seed` propagates seeds inside `mclapply`.
- Per-replicate seeds are `master_seed + rep_idx`.

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.

Each simulation driver sets `set.seed(20260507)` at the top of the replicate loop and uses `parallel::mc.set.seed` inside `mclapply`. Per-replicate seeds derive from the master seed by `+ rep_idx`.

[96]

All simulation outputs, driver scripts, pre-registration, and manuscript source files are versioned at documented paths within the repository.

- Per-replicate outputs and Morris summaries are at `analysis/data/quick-sim/`.
- Driver scripts and the ADEMP pre-registration are at `analysis/scripts/component-decomposition/`.
- Manuscript source, bibliography, and CSL file are at `analysis/report/06-component-decomposition/`.

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.

Per-replicate simulation outputs are checkpointed at `analysis/data/quick-sim/06-decomp-replicates`.
Morris ADEMP cell-level summaries at the companion `06-decomp-summary.txt`.
Driver scripts are under `analysis/scripts/component-decomposition/`.
The pre-registered ADEMP plan is at `analysis/scripts/component-decomposition/00-ademp-pre-reg`.
The manuscript source, paper-specific bibliography (`references.bib`),
and SIM CSL file are at `analysis/report/06-component-decomposition/`.

15 References

1. Holford N. Clinical pharmacology = disease progression + drug action. *British Journal of Clinical Pharmacology*. 2015;79(1):18-27. doi:10.1111/bcp.12170
2. Gomeni R, Lavergne A, Merlo-Pich E. Modelling placebo response in depression trials using a longitudinal model with informative dropout. *European Journal of Pharmaceutical Sciences*. 2009;36(1):4-10. doi:10.1016/j.ejps.2008.10.025

- 2677 3. Chen C, Jönsson S, Yang S, Plan EL, Karlsson MO. Detecting placebo and drug effects on Parkinson's disease symptoms by longitudinal item-score models. *CPT: Pharmacometrics & Systems Pharmacology*. 2021;10(4):309-317. doi:10.1002/psp4.12601
- 2678 4. Muthén B, Brown HC. Estimating drug effects in the presence of placebo response: Causal inference using growth mixture modeling. *Statistics in Medicine*. 2009;28(27):3363-3385. doi:10.1002/sim.3721
- 2679 5. Kaptchuk TJ, Kelley JM, Conboy LA, et al. Components of placebo effect: Randomised controlled trial in patients with irritable bowel syndrome. *BMJ*. 2008;336(7651):999-1003. doi:10.1136/bmj.39524.439618.25
- 2680 6. Kirsch I, Deacon BJ, Huedo-Medina TB, Scoboria A, Moore TJ, Johnson BT. Initial severity and antidepressant benefits: A meta-analysis of data submitted to the Food and Drug Administration. *PLoS Medicine*. 2008;5(2):e45. doi:10.1371/journal.pmed.0050045
- 2681 7. Locher C, Kossowsky J, Gaab J, Kirsch I, et al. Moderation of antidepressant and placebo outcomes by baseline severity in late-life depression: A systematic review and meta-analysis. *Journal of Affective Disorders*. 2015;181:50-60. doi:10.1016/j.jad.2015.03.062
- 2682 8. Sugarman MA, Loree AM, Baltes BB, Grekin ER, Kirsch I. The efficacy of paroxetine and placebo in treating anxiety and depression. *PLoS ONE*. 2014;9(8):e106337. doi:10.1371/journal.pone.0106337
- 2683 9. Hall KT, Lembo AJ, Kirsch I, et al. Catechol-O-methyltransferase val158met polymorphism predicts placebo effect in irritable bowel syndrome. *PLoS ONE*. 2012;7(10):e48135. doi:10.1371/journal.pone.0048135
- 2684 10. Wang RS, Lembo AJ, Kaptchuk TJ, Hall KT, et al. Genomic effects associated with response to placebo treatment in a randomized trial of irritable bowel syndrome. *Frontiers in Pain Research*. 2022;2:775386. doi:10.3389/fpain.2021.775386
- 2685 11. Hróbjartsson A, Gøtzsche PC. Is the placebo powerless? An analysis of clinical trials comparing placebo with no treatment. *New England Journal of Medicine*. 2001;344(21):1594-1602. doi:10.1056/NEJM200105243442106
- 2686 12. Hróbjartsson A, Gøtzsche PC. Is the placebo powerless? Update of a systematic review with 52 new randomized trials. *Journal of Internal Medicine*. 2004;256(2):91-100. doi:10.1111/j.1365-2796.2004.01355.x

- 2687 13. Hróbjartsson A, Gøtzsche PC. Placebo interventions for all clinical conditions. *Cochrane Database of Systematic Reviews*. 2010;(1):CD003974. doi:10.1002/14651858.CD003974.pub3
- 2688 14. Kent DM, Paulus JK, Klaveren D van, et al. The Predictive Approaches to Treatment effect Heterogeneity (PATH) Statement. *Annals of Internal Medicine*. 2020;172(1):35-45. doi:10.7326/M18-3667
- 2689 15. Kent DM, Klaveren D van, Paulus JK, et al. The PATH statement: Explanation and elaboration. *Annals of Internal Medicine*. 2020;172(1):W1-W25. doi:10.7326/M18-3668
- 2690 16. Kent DM, Steyerberg E, Klaveren D van. Personalized evidence based medicine: Predictive approaches to heterogeneous treatment effects. *BMJ*. 2018;363:k4245. doi:10.1136/bmj.k4245
- 2691 17. Rekkas A, Paulus JK, Raman G, et al. Predictive approaches to heterogeneous treatment effects: A scoping review. *BMC Medical Research Methodology*. 2020;20(1):264. doi:10.1186/s12874-020-01145-1
- 2692 18. Guyatt G, Sackett D, Taylor DW, Chong J, Roberts R, Pugsley S. Determining optimal therapy: Randomized trials in individual patients. *New England Journal of Medicine*. 1986;314(14):889-892.
- 2693 19. Zucker DR, Schmid CH, McIntosh MW, D'Agostino RB, Selker HP, Lau J. Combining single patient (N-of-1) trials to estimate population treatment effects and to evaluate individual patient responses to treatment. *Journal of Clinical Epidemiology*. 1997;50(4):401-410. doi:10.1016/S0895-4356(96)00429-5
- 2694 20. Zucker DR, Ruthazer R, Schmid CH. Individual (N-of-1) trials can be combined to give population comparative treatment effect estimates: Methodologic considerations. *Journal of Clinical Epidemiology*. 2010;63(12):1312-1323. doi:10.1016/j.jclinepi.2010.04.020
- 2695 21. Lillie EO, Patay B, Diamant J, Issell B, Topol EJ, Schork NJ. The n-of-1 clinical trial: The ultimate strategy for individualizing medicine? *Personalized Medicine*. 2011;8(2):161-173. doi:10.2217/pme.11.7
- 2696 22. Duan N, Kravitz RL, Schmid CH. Single-patient (n-of-1) trials: A pragmatic clinical decision methodology for patient-centered comparative effectiveness research. *Journal of Clinical Epidemiology*. 2013;66(8 Suppl):S21-S28. doi:10.1016/j.jclinepi.2013.04.006

- 2697 23. Gabler NB, Duan N, Vohra S, Kravitz RL. N-of-1 trials in the medical literature: A systematic review. *Medical Care*. 2011;49(8):761-768. doi:10.1097/MLR.0b013e318215d90d
- 2698 24. Schork NJ, Goetz LH. Single-subject studies in translational nutrition research. *Annual Review of Nutrition*. 2017;37:395-422. doi:10.1146/annurev-nutr-071816-064717
- 2699 25. Schork NJ. Randomized clinical trials and personalized medicine: A commentary on Deaton and Cartwright. *Social Science and Medicine*. 2018;210:71-73. doi:10.1016/j.socscimed.2018.04.033
- 2700 26. Schork NJ, Beaulieu-Jones B, Liang WS, Smalley S, Goetz LH. Exploring human biology with N-of-1 clinical trials. *Cambridge Prisms: Precision Medicine*. 2023;1:e12. doi:10.1017/pcm.2022.15
- 2701 27. Porcino A, Vohra S. N-of-1 trials, their reporting guidelines, and the advancement of open science principles. *Harvard Data Science Review*. 2022;4(SI3). doi:10.1162/99608f92.a65a257a
- 2702 28. Vohra S, Shamseer L, Sampson M, et al. CONSORT extension for reporting N-of-1 trials (CENT) 2015 statement. *Journal of Clinical Epidemiology*. 2015;76:9-17. doi:10.1016/j.jclinepi.2015.05.004
- 2703 29. Shamseer L, Sampson M, Bukutu C, Schmid CH, et al. CONSORT extension for reporting N-of-1 trials (CENT) 2015: Explanation and elaboration. *Journal of Clinical Epidemiology*. 2015;76:18-46. doi:10.1016/j.jclinepi.2015.05.018
- 2704 30. Li J, Gao W, Punja S, et al. Reporting quality of N-of-1 trials published between 1985 and 2013: A systematic review. *Journal of Clinical Epidemiology*. 2016;76:57-64. doi:10.1016/j.jclinepi.2015.11.016
- 2705 31. Liao Z, Qian M, Kronish IM, Cheung YK. Analysis of N-of-1 trials using Bayesian distributed lag model with autocorrelated errors. *Statistics in Medicine*. 2023;42(13):2044-2060. doi:10.1002/sim.9676
- 2706 32. Blackston JW, Chapple AG, McGree JM, McDonald S, Nikles J. Comparison of aggregated N-of-1 trials with parallel and crossover randomized controlled trials using simulation studies. *Healthcare*. 2019;7(4):137. doi:10.3390/healthcare7040137

- 2707 33. Carrozzo AE, Zimmermann G, Bathke AC, et al. Two-arm crossover randomized controlled trial versus meta-analysis of N-of-1 studies: Comparison of statistical efficiency. *Biometrical Journal*. 2025;67(2):e70045. doi:10.1002/bimj.70045
- 2708 34. Hendrickson RC, Thomas RG, Schork NJ, Raskind MA. Optimizing aggregated N-of-1 trial designs for predictive biomarker validation: Statistical methods and practical considerations. *Frontiers in Digital Health*. 2020;2:13. doi:10.3389/fdgth.2020.00013
- 2709 35. Kaptchuk TJ, Friedlander E, Kelley JM, et al. Placebos without deception: A randomized controlled trial in irritable bowel syndrome. *PLoS ONE*. 2010;5(12):e15591. doi:10.1371/journal.pone.0015591
- 2710 36. Lembo A, Kelley JM, Nee J, et al. Open-label placebo vs double-blind placebo for irritable bowel syndrome: A randomized clinical trial. *Pain*. 2021;162(9):2428-2435. doi:10.1097/j.pain.0000000000002234
- 2711 37. Nurko S, Saps M, Kossowsky J, others, Kelley JM. Effect of open-label placebo on children and adolescents with functional abdominal pain or irritable bowel syndrome. *JAMA Pediatrics*. 2022;176(4):349-356. doi:10.1001/jamapediatrics.2021.5750
- 2712 38. Benedetti F. Placebo effects: From the neurobiological paradigm to translational implications. *Neuron*. 2014;84(3):623-637.
- 2713 39. Frisaldi E, Shaibani A, Benedetti F. Understanding the mechanisms of placebo and nocebo effects. *Swiss Medical Weekly*. 2020;150:w20340. doi:10.4414/smw.2020.20340
- 2714 40. Carlino E, Frisaldi E, Benedetti F. Pain and the context. *Nature Reviews Rheumatology*. 2014;10(6):348-355. doi:10.1038/nrrheum.2014.17
- 2715 41. Carlino E, Piedimonte A, Benedetti F. Nature of the placebo and nocebo effect in relation to functional neurologic disorders. In: *Handbook of Clinical Neurology*. Vol 139.; 2016:597-606. doi:10.1016/B978-0-12-801772-2.00048-5
- 2716 42. Wager TD, Atlas LY, Lindquist MA, Roy M, Woo CW, Kross E. An fMRI-based neurologic signature of physical pain. *New England Journal of Medicine*. 2013;368(15):1388-1397. doi:10.1056/NEJMoa1204471

- 2717 43. Zunhammer M, Bingel U, Wager TD. Placebo effects on the neurologic pain signature: A meta-analysis of individual participant functional magnetic resonance imaging data. *JAMA Neurology*. 2018;75(11):1321-1330. doi:10.1001/jamaneurol.2018.2017
- 2718 44. Raskind MA, Peterson K, Williams T, et al. A trial of prazosin for combat trauma PTSD with nightmares in active-duty soldiers returned from Iraq and Afghanistan. *American Journal of Psychiatry*. 2013;170(9):1003-1010. doi:10.1176/appi.ajp.2013.12081133
- 2719 45. Holford NHG. Holford NHG and sheiner LB 'understanding the dose-effect relationship - clinical application of pharmacokinetic-pharmacodynamic models', Clin Pharmacokin 6:429-453 (1981) - the backstory. *The AAPS Journal*. 2011;13(4):662-664. doi:10.1208/s12248-011-9306-5
- 2720 46. Tilburt JC, Emanuel EJ, Kaptchuk TJ, Curlin FA, Miller FG. Prescribing 'placebo treatments': Results of national survey of US internists and rheumatologists. *BMJ*. 2008;337:a1938. doi:10.1136/bmj.a1938
- 2721 47. Finniss DG, Kaptchuk TJ, Miller F, Benedetti F. Biological, clinical, and ethical advances of placebo effects. *The Lancet*. 2010;375(9715):686-695. doi:10.1016/S0140-6736(09)61706-2
- 2722 48. Kirsch I. Response expectancy as a determinant of experience and behavior. *American Psychologist*. 1985;40(11):1189-1202. doi:10.1037/0003-066X.40.11.1189
- 2723 49. Senn S. Controversies concerning randomization and additivity in clinical trials. *Statistics in Medicine*. 2004;23(24):3729-3753. doi:10.1002/sim.2074
- 2724 50. Bingel U, Wanigasekera V, Wiech K, et al. The effect of treatment expectation on drug efficacy: Imaging the analgesic benefit of the opioid remifentanyl. *Science Translational Medicine*. 2011;3(70):70ra14. doi:10.1126/scitranslmed.3001244
- 2725 51. Colloca L, Lopiano L, Lanotte M, Benedetti F. Overt versus covert treatment for pain, anxiety, and Parkinson's disease. *The Lancet Neurology*. 2004;3(11):679-684.
- 2726 52. Rohsenow DJ, Marlatt GA. The balanced placebo design: Methodological considerations. *Addictive Behaviors*. 1981;6(2):107-122.

- 2727 53. International Council for Harmonisation of Technical Requirements for
Pharmaceuticals for Human Use. ICH E9(R1) addendum on estimands
and sensitivity analysis in clinical trials to the guideline on statistical prin-
ciples for clinical trials. Published online 2019.
- 2728 54. Senn S. Statistical pitfalls of personalized medicine. *Nature*.
2018;563(7733):619-621.
- 2729 55. Daza EJ. Causal analysis of self-tracked time series data using a counter-
factual framework for N-of-1 trials. *Methods of Information in Medicine*.
2018;57(S 01):e10-e21. doi:10.3414/ME16-02-0044
- 2730 56. Gärtner T, Schneider J, Arnrich B, Konigorski S. Comparison of Bayesian
networks, G-estimation and linear models to estimate causal treatment
effects in aggregated N-of-1 trials with carry-over effects. *BMC Medical
Research Methodology*. 2023;23:191. doi:10.1186/s12874-023-02012-5
- 2731 57. Morris TP, White IR, Crowther MJ. Using simulation studies to evaluate
statistical methods. *Statistics in Medicine*. 2019;38(11):2074-2102.